

# Supplementary Material

## Evaluation area and the limits of cross-region transfer of pre-fire thermal wildfire models: five large Mediterranean fires

Supplementary material for *Evaluation area and the limits of cross-region transfer of pre-fire thermal wildfire models: five large Mediterranean fires*.

Section S1 gives the sensitivity analyses and the details behind the Results. Section S2 gives the supporting tables, Section S3 the protocol details and limitations, Section S4 the target-label recovery curve, Section S5 the code and Section S6 the data. Tables are numbered S1 to S36, independently of the sections. Supplementary equations are numbered (S1), (S2) and so on. Every value is read from a released output, named by its path in the repository <https://github.com/dryuemco/thermal-twin>. Paths that begin with paper/labelfix\_rerun/ hold the corrected-label outputs. Values computed under the original Manavgat label are marked “original label”.

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## 44   **S1 Sensitivity analyses and detail**

45   Each analysis below changes one design choice and keeps everything else fixed. None  
 46   changes a conclusion of the main text. Section S1.6 is different: it tests a claim of the  
 47   introduction, and the claim is not supported.

## **S1.1 Evia AOI and prevalence**

The raw transfer analyses between Evia and Bejís, Manavgat and Muğla were repeated with the earlier, high-prevalence Evia rectangle. No qualitative conclusion changed. Thermal raw transfer AUCs moved by up to 0.134. The largest change was Evia to Manavgat, at 0.543 with the earlier rectangle against 0.677 with the extended one. No direction changed side of the chance line (paper/labelfix\_rerun/round7/r7f\_evia\_legacy.csv). On the earlier rectangle the natural-vegetation population is smaller, and its burned prevalence is higher than on the extended rectangle (0.287; Table S15).

## **S1.2 CORAL regularisation**

Nine  $\lambda$  values from 0 to  $10^{-1}$  were tested on four directions (Bejís and Muğla, and Manavgat and Muğla, both ways). CORAL transfer AUC moved by at most 0.014 within any direction and model, and by at most 0.009 for the thermal model (paper/labelfix\_rerun/exports/coral\_lambda\_sensitivity\_b74d643e/metrics.csv).  $\lambda$  was not selected on performance.

The Manavgat and Bejís pair is not in this sweep. Under the original label, the adapted result of this pair depended on  $\lambda$ . It was therefore tested separately at  $\lambda = 10^{-5}$ ,  $10^{-3}$ ,  $10^{-1}$  and 1 with the authors' own re-implementation of the pipeline's CORAL (paper/labelfix\_rerun/step10/coral\_lambda\_sensitivity.csv). This implementation differs slightly from the pipeline's: at  $\lambda = 10^{-5}$  it gives 0.408 for Bejís to Manavgat and 0.470 for Manavgat to Bejís, against 0.406 and 0.467 in Table S16. Across the four  $\lambda$  values, Bejís to Manavgat gives 0.408 [0.389, 0.427], 0.410, 0.367 and 0.313, which are below chance at every  $\lambda$ . Manavgat to Bejís gives 0.470 [0.444, 0.494], 0.474, 0.466 and 0.457. No adapted direction of this pair has an interval above chance at any  $\lambda$ , so no conclusion depends on the

choice of  $\lambda$ . The third implementation, of Section S1.23 (R8c), reproduces the pipeline's CORAL matrix to within 0.009.

### S1.3 Blocking scale

The transfer intervals were recomputed at 10-cell ( $\approx 5$  km) blocking from the re-frozen per-cell predictions. The intervals became wider. The paired-delta verdicts changed from twelve positive, seven negative and one uncertain at 1 km to six, five and nine at 5 km. Coarser blocking therefore removed support from eight verdicts and added none. Across five bootstrap seeds, the 5 km counts were 5 to 6 positive, 4 to 5 negative and 9 to 11 uncertain, so they are not exact (paper/labelfix\_rerun/round5/out\_official/transfer\_ci\_blocksize.csv, paper/labelfix\_rerun/round6/seed\_stability.json).

The point estimates do not change, because the block size only sets the bootstrap resampling unit. Each point estimate is computed once over all target cells. Every verdict that changed moved to “no verdict”, and no direction crossed the chance line. The counts of supported directions are therefore the least stable numbers in this paper.

**Table 1 note (resampling units).** The bootstrap resamples spatial blocks, so the reliability of an interval depends on the number of blocks with at least one burned cell. This number falls from 192 to 843 at 2 cells to **6 to 33** at 20 cells. An interval built on six such blocks has no useful coverage, so **the 20-cell row should be read as indicative**. The 10-cell row is the coarsest blocking that this design supports. There, every region has 16 to 70 blocks with burned cells, and the gain holds in all five regions.

### S1.4 Predictor-window closure

The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stayed positive, with bootstrap support, everywhere. The direction of the change

differed between regions. In Manavgat, on the corrected label, it was +0.065 [+0.059, +0.072] at the standard window, +0.082 [+0.075, +0.090] at 7 days and +0.058 [+0.051, +0.065] at 14 days, all on the common cohort of that analysis (paper/labelfix\_rerun/exports/window\_closure\_manavgat\_2021/). It became stronger in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It was flat in Montiferru. It became weaker in Evia, from 0.156 to 0.149 to 0.135. The four regions other than Manavgat are not affected by the label correction (paper/labelfix\_rerun/exports/window\_closure\_<region>/tables/thermal\_contributions.csv). The gain therefore survives in every region, but it does not always improve.

### **S1.5 Quality screening of the coarse thermal input**

The MODIS inputs were quality-screened in two regions and not in the other three. This split follows the export date, not the design. It causes a change at the input that is correlated with elevation ( $r = +0.615$  in Manavgat). The Manavgat chain was therefore rebuilt from a quality-screened input on the corrected label and compared with the unscreened version. The current pipeline step7 refuses the unscreened raster, because the raster has no nodata tag and 8.1 % exact zeros. The unscreened version therefore uses the step7 of export time, and the screened version uses the current one. The two versions differ in code as well as in screening. The result did not change. Elevation stayed at 0.232, no other signed AUC moved by more than 0.005 (downscaled LST,  $-0.0044$ ), and the within-region gain moved from +0.067 to +0.068 (paper/labelfix\_rerun/round6/qc/qc\_compare\_manavgat\_corrected.json; Section S3.5(xii)). Elevation comes from the DEM and is not affected by the screening. The MODIS-based surface is used for only 2.14 percentage points of coverage. The Muğla input was not re-examined.

## S1.6 Normalised against absolute dryness channels

Section 1.2 states that an internally normalised index should be less affected than raw land surface temperature by temperature differences between regions. The thermal set contains both kinds of channels, so this was tested directly by comparing feature sets.

In the two Muğla fires (Section S1.13), the normalised channels keep their direction and the absolute channels do not. This holds for one region across two fires. It does not hold across regions: `1st_anomaly_mean` is one of the predictors with a bootstrap-supported reversal on the original study areas (Table S10). For this reason, Section S1.14 removes it together with elevation. On this evidence, normalisation protects a channel against a difference between two seasons in one place, but not against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds. The classifier, population, folds and bootstrap were kept fixed. The code stops unless its reference configuration reproduces the re-frozen exports, and it does: the largest absolute difference from the step9b transfer AUCs is 0.000000 across all twenty directions. Sources for this section and the next two: `paper/labelfix_rerun/round3/no_coord_channels.json` and `paper/labelfix_rerun/code/model_capacity.json`, summarised by `paper/labelfix_rerun/round7/r7e_supplement_tables.py`.

**Table S23. Feature-set arms.** Mean over the twenty transfer directions and the five regions.

| Feature set                       | Mean transfer<br>AUC | Directions > 0.5 | Mean within-region<br>AUC |
|-----------------------------------|----------------------|------------------|---------------------------|
| Baseline only                     | 0.519                | 13               | 0.797                     |
| Baseline + normalised anomalies   | 0.524                | 14               | 0.862                     |
| Baseline + absolute surface state | 0.531                | 13               | 0.864                     |
| All ten features (reference)      | 0.527                | 13               | 0.896                     |

**The expected advantage was not found.** Per direction, the normalised set minus the absolute set has a mean of  $-0.006$ . The difference is positive in 11 of 20 directions and ranges from

−0.099 to +0.087, so the spread is much larger than the mean difference. Within regions the two sets are almost equal, at 0.862 and 0.864. Each recovers most of the gap between the baseline (0.797) and the full model (0.896). The comparison uses two parts of one thermal set on one cohort. It does not test normalised dryness indices in general.

## S1.7 The coordinate-informed channels

downscaled\_1st\_mean and fused\_1st\_mean come from a per-region downscaling model that uses coordinates as inputs (Section S3.4). This is the only way in which coordinate information enters the feature set. A surface smoothed by coordinates is informative locally and does not transfer. If the within-region gain depended on it, the gain would be caused by the smoothing and not by the thermal state. Both channels were therefore removed, and all models were refitted over all twenty directions and all five within-region folds.

**Table S24. The within-region increment without the two coordinate-informed channels.**

| Region          | Increment, full set | Without the two channels | Retained    |
|-----------------|---------------------|--------------------------|-------------|
| Manavgat 2021   | +0.067              | +0.064                   | 95 %        |
| Bejís 2022      | +0.056              | +0.046                   | 82 %        |
| Muğla 2021      | +0.116              | +0.097                   | 84 %        |
| North Evia 2021 | +0.153              | +0.145                   | 94 %        |
| Montiferru 2021 | +0.101              | +0.105                   | 103 %       |
| <b>Mean</b>     | <b>+0.099</b>       | <b>+0.091</b>            | <b>92 %</b> |

**The gain does not depend on these channels.** It stays positive in every region and keeps 82 % to 103 % of its size. Mean transfer also does not change: 0.528 without the two channels, 0.527 with them, and 0.519 for the baseline alone.

## S1.8 Model capacity

All numbers in the paper come from one random forest with unlimited depth and `min_samples_leaf = 3`. This configuration fits local structure well and extrapolates poorly.

Three other estimators were therefore run over the same twenty directions and five within-region folds.

**Table S25. Four estimators, within region and in transfer.**

| Estimator                              | Within-region<br>AUC | Within<br>increment | Transfer<br>AUC | Transfer<br>increment | Above<br>chance |
|----------------------------------------|----------------------|---------------------|-----------------|-----------------------|-----------------|
| Random forest, depth unlimited, leaf 3 | 0.896                | +0.099              | 0.527           | +0.007                | 13 of 20        |
| Random forest, depth 6, leaf 50        | 0.838                | +0.053              | 0.527           | −0.011                | 13 of 20        |
| Random forest, leaf 200                | 0.814                | +0.046              | 0.522           | −0.019                | 13 of 20        |
| Penalised logistic regression          | 0.759                | +0.046              | 0.481           | −0.021                | 11 of 20        |

**Regularisation does not improve transfer.** The four estimators range from 0.481 to 0.527, with eleven to thirteen of twenty directions above chance. The linear model, which is designed to extrapolate, transfers worst. Within-region skill falls from 0.896 to 0.759 as capacity is reduced, but transfer does not rise. The thermal contribution to transfer is +0.007 for the standard forest and negative for all three alternatives. The estimators are not ranked, because only point estimates are available.

### S1.9 The four evaluations of Section 4.3, in full

Section 4.3 reports four evaluations of the same models (main-text Table 2). The values per scar, the controls and the thermal gain per scar are given here.

**Table S1. The four evaluations, per scar.** Natural-vegetation population, 5 km blocking, 2 km collar. Bejís and Manavgat have no row C, because each has only one burned area, and withholding it leaves nothing to train on. Source [paper/labelfix\\_rerun/code/matched\\_holdout.json](#).

| Region        | Scar | Cells | Burned fraction | A whole<br>region | B scar area | C scar withheld | D<br>foreign |
|---------------|------|-------|-----------------|-------------------|-------------|-----------------|--------------|
| Manavgat 2021 | 1    | 3,965 | 0.74            | 0.882             | 0.743       | n/a             | 0.542        |
| Bejís 2022    | 1    | 1,657 | 0.66            | 0.824             | 0.575       | n/a             | 0.589        |
| Muğla 2021    | 6    | 1,266 | 0.72            | 0.777             | 0.564       | 0.595           | 0.571        |
| Muğla 2021    | 1    | 1,244 | 0.59            | 0.777             | 0.654       | 0.561           | 0.514        |
| Muğla 2021    | 10   | 940   | 0.68            | 0.777             | 0.674       | 0.617           | 0.634        |

| Region          | Scar | Cells | Burned fraction | A whole region | B scar area | C scar withheld | D foreign |
|-----------------|------|-------|-----------------|----------------|-------------|-----------------|-----------|
| Muğla 2021      | 8    | 954   | 0.57            | 0.777          | 0.761       | 0.542           | 0.623     |
| North Evia 2021 | 1    | 3,059 | 0.87            | 0.864          | 0.745       | 0.465           | 0.564     |
| Montiferru 2021 | 1    | 758   | 0.58            | 0.720          | 0.622       | 0.584           | 0.489     |
| Montiferru 2021 | 5    | 195   | 0.34            | 0.720          | 0.461       | 0.458           | 0.475     |

### 171 **S1.9.1 The pool controls**

172 The same predictions were scored on a random sample of region cells with the burned fraction  
173 of the scar area. Over nine scars this gives 0.793, against 0.791 for the whole region.

174 Matching the prevalence therefore changes nothing ( $-0.002$  [ $-0.005$ ,  $+0.001$ ]). This is  
175 expected, because ROC-AUC does not depend on class balance, so it is a check of the code.

176 Scoring the predictions on the scar area gives 0.644, a fall of **+0.149** [**+0.087**, **+0.211**] from  
177 the prevalence-matched score. This replaces both the burned and the unburned cells, so each  
178 group was also replaced alone. The cost is always the prevalence-matched score minus the  
179 new score. Replacing only the unburned cells costs **0.139** [**0.098**, **0.181**] (0.046 to 0.250 per  
180 scar). Replacing only the burned cells costs **-0.002** [**-0.055**, **+0.051**] on average ( $-0.121$  to  
181  $+0.137$  per scar). The cost therefore comes from the unburned cells. All unburned cells in a  
182 scar collar are next to the fire and share its terrain, land cover and weather. The unburned cells  
183 of a whole region also include easy distant cells  
184 (`paper/labelfix_rerun/code/pool_decomposition.json`, `prevalence_control.json`).

### 185 **S1.9.2 The seven scars and the resampling unit**

186 Rows A to D of Table 2 are means over the **same seven scars**: four in Muğla, two in  
187 Montiferru and one in Evia. Under the corrected label, the 2,151 added Manavgat burned cells  
188 join its one existing scar (2,934 of 2,935 cells). For this reason the controls above use nine  
189 scars and report 0.791 and 0.644, against 0.773 and 0.640 in Table 2. The intervals are  
190 Student  $t$  intervals over the scars. **Row A is a region-level value that is repeated for each**

scar of a region, so its interval is too narrow and should not be read as coverage. The same applies to  $A - B$  and  $A - C$ . With the region as the unit,  $A - B$  is **+0.137 [+0.048, +0.226]** over three regions and **+0.160 [+0.090, +0.230]** over all nine scars in five regions (paper/labelfix\_rerun/round7/round7\_summary.json).  $A - C$  with the region as the unit is **+0.266 [-0.022, +0.553]**. This interval is 3.6 times wider than the scar-level interval and includes zero. **With the region as the unit, only the frame cost is established.**

The scar definition does not drive the result. Over a range of minimum scar sizes and buffers, row C ranges from 0.541 to 0.561 on nine to six scars. Its means at 2, 5 and 10 km buffers are 0.546 (seven scars), 0.533 (seven) and 0.548 (six) (paper/labelfix\_rerun/code/scar\_definition\_sweep.csv, scar\_control.json). For single scars, row D varies strongly between sources (Table S4), but little after averaging. Row D uses 28 of the 36 source and scar combinations, and nine of them are below chance.

### S1.9.3 Controls and the thermal-increment ladder, full specification

Three controls reuse the predictions of row A. Each is averaged over 20 draws without replacement. The **prevalence-matched** control draws  $|H_K^+|$  cells from  $P_R$  and  $|H_K^-|$  cells from  $V_R \setminus P_R$ . The **negative-pool** control keeps the drawn burned cells but uses the unburned cells of the scar,  $H_K^-$ . The positive-pool control does the opposite. A **within-region half-split** completes the set: the modelled cells are cut at the median of a grid axis, in both axes and both directions, and the transfer protocol of Section 3.8 is applied. A split is dropped when one half has only one class. The numbers of burned cells in the two halves are unequal (Table S2).

The **thermal gain** is evaluated with rows A to D on the seven scars of Table 2. Table S26 gives the values per scar at both block sizes. Over the seven scars, the region-wide gain is **+0.095** at 5 km blocking and **+0.117** at 1 km. On the scar area, the same predictions give

215 +0.021 and +0.056. Leave-one-scar-out gives +0.024, and a foreign-region model +0.008, at  
 216 both block sizes. Blocked minus leave-one-scar-out is  $-0.004$   $[-0.070, +0.063]$  at 5 km and  
 217  $+0.031$   $[-0.027, +0.090]$  at 1 km. Region-wide minus scar area at 5 km is  $+0.074$   $[-0.009,$   
 218  $+0.157]$  at scar level. With only three regions, no reliable region-level interval can be given  
 219 for this difference. The within-region half-split gives +0.028, positive in 13 of 18 splits  
 220 (paper/labelfix\_rerun/inference/ladder\_summary.json,  
 221 paper/labelfix\_rerun/code/positive\_control.json).

222 **Table S26. The thermal gain by scar: region-wide, on the scar area, withheld and**  
 223 **foreign.** Thermal minus baseline ROC-AUC. Region-wide and scar-area values at 5 km  
 224 blocking, with 1 km in brackets. Leave-one-scar-out and foreign values do not depend on the  
 225 blocking.

| Region          | Scar | Region-wide            | Scar frame, same blocked model | Scar withheld | Foreign       |
|-----------------|------|------------------------|--------------------------------|---------------|---------------|
| Muğla 2021      | 6    | +0.079 (+0.116)        | $-0.016$ (+0.018)              | +0.025        | +0.033        |
| Muğla 2021      | 1    | +0.079 (+0.116)        | +0.049 (+0.071)                | $-0.044$      | +0.014        |
| Muğla 2021      | 10   | +0.079 (+0.116)        | +0.063 (+0.107)                | +0.045        | +0.042        |
| Muğla 2021      | 8    | +0.079 (+0.116)        | +0.083 (+0.092)                | +0.012        | +0.037        |
| North Evia 2021 | 1    | +0.148 (+0.153)        | +0.060 (+0.065)                | +0.143        | $-0.038$      |
| Montiferru 2021 | 1    | +0.100 (+0.101)        | +0.066 (+0.086)                | +0.060        | $-0.008$      |
| Montiferru 2021 | 5    | +0.100 (+0.101)        | $-0.160$ ( $-0.049$ )          | $-0.070$      | $-0.023$      |
| <b>Mean</b>     |      | <b>+0.095 (+0.117)</b> | <b>+0.021 (+0.056)</b>         | <b>+0.024</b> | <b>+0.008</b> |

226 **Table S2. Within-region half-split, every split.** The numbers of burned cells in the source  
 227 and target halves are given because they are unequal. This is the main limit of this analysis,  
 228 because a straight cut does not give two comparable halves. Two splits cannot be used,  
 229 because one half of Manavgat has no burned cells. Source  
 230 paper/labelfix\_rerun/code/positive\_control.json; checked row by row by  
 231 paper/code/appendix\_tables.py.

| Region        | Axis      | Direction   | Source positives | Target positives | Thermal AUC | Baseline AUC |
|---------------|-----------|-------------|------------------|------------------|-------------|--------------|
| manavgat 2021 | east-west | low to high | 1,795            | 1,140            | 0.761       | 0.793        |

| Region             | Axis        | Direction   | Source positives | Target positives | Thermal AUC | Baseline AUC |
|--------------------|-------------|-------------|------------------|------------------|-------------|--------------|
| manavgat 2021      | east-west   | high to low | 1,140            | 1,795            | 0.712       | 0.638        |
| manavgat 2021      | north-south | low to high | 0                | 2,935            | n/a         | n/a          |
| manavgat 2021      | north-south | high to low | 2,935            | 0                | n/a         | n/a          |
| bejis 2022         | east-west   | low to high | 640              | 460              | 0.750       | 0.784        |
| bejis 2022         | east-west   | high to low | 460              | 640              | 0.578       | 0.560        |
| bejis 2022         | north-south | low to high | 357              | 743              | 0.713       | 0.662        |
| bejis 2022         | north-south | high to low | 743              | 357              | 0.603       | 0.613        |
| mugla 2021         | east-west   | low to high | 1,449            | 1,462            | 0.448       | 0.378        |
| mugla 2021         | east-west   | high to low | 1,462            | 1,449            | 0.636       | 0.628        |
| mugla 2021         | north-south | low to high | 774              | 2,137            | 0.533       | 0.503        |
| mugla 2021         | north-south | high to low | 2,137            | 774              | 0.294       | 0.549        |
| evia 2021 extended | east-west   | low to high | 2,052            | 612              | 0.589       | 0.506        |
| evia 2021 extended | east-west   | high to low | 612              | 2,052            | 0.645       | 0.482        |
| evia 2021 extended | north-south | low to high | 2,564            | 100              | 0.649       | 0.513        |
| evia 2021 extended | north-south | high to low | 100              | 2,564            | 0.533       | 0.495        |
| montiferru 2021    | east-west   | low to high | 410              | 129              | 0.475       | 0.461        |
| montiferru 2021    | east-west   | high to low | 129              | 410              | 0.507       | 0.513        |
| montiferru 2021    | north-south | low to high | 438              | 101              | 0.583       | 0.511        |
| montiferru 2021    | north-south | high to low | 101              | 438              | 0.536       | 0.447        |

**Table S3. Leave-one-scar-out at a 2 km buffer, every scar.** Only Muğla and Montiferru contain more than one burned area of at least 50 cells. Only in Muğla does every held-out case still leave a well-trained source model. The two Montiferru areas are very unequal, so one case keeps 472 burned source cells and the other 97. The mean is 0.579 for the four Muğla cases and 0.502 for the three others. The pooled mean, 0.546, is row C of Table 2.

237 Source paper/labelfix\_rerun/code/scar\_control.json; checked row by row by  
 238 paper/code/appendix\_tables.py.

| Region             | Component | Source positives left | Target positives | Target cells | AUC   |
|--------------------|-----------|-----------------------|------------------|--------------|-------|
| evia 2021 extended | 1         | 11                    | 2,653            | 3,059        | 0.465 |
| montiferru 2021    | 1         | 97                    | 442              | 758          | 0.584 |
| montiferru 2021    | 5         | 472                   | 67               | 195          | 0.458 |
| mugla 2021         | 6         | 1,997                 | 914              | 1,266        | 0.595 |
| mugla 2021         | 1         | 2,173                 | 738              | 1,244        | 0.561 |
| mugla 2021         | 10        | 2,272                 | 639              | 940          | 0.617 |
| mugla 2021         | 8         | 2,363                 | 548              | 954          | 0.542 |

239 The evaluation populations of these analyses cannot be compared with each other or with the  
 240 transfer targets. A held-out scar with its 2 km collar contains only unburned cells next to the  
 241 fire, while a whole target region also contains easy distant cells. The burned fractions (34 to  
 242 87 %, against 7.0 to 28.7 %) are a result of this difference, not its cause.

243 **Table S4. The foreign-region evaluation, by source.** Each held-out scar area is scored with  
 244 a model fitted on each of the other four regions. Row D of Table 2 is the mean over the seven  
 245 scars that have a row C (28 of the 36 combinations below). Over all nine scars the mean is  
 246 0.556. Source paper/labelfix\_rerun/code/d\_per\_source.json; checked row by row by  
 247 paper/code/appendix\_tables.py.

| Target region   | Scar | Mean over sources | Min   | Max   | Spread |
|-----------------|------|-------------------|-------|-------|--------|
| Bejís 2022      | 1    | 0.589             | 0.519 | 0.631 | 0.112  |
| North Evia 2021 | 1    | 0.564             | 0.374 | 0.704 | 0.329  |
| Manavgat 2021   | 1    | 0.542             | 0.438 | 0.665 | 0.227  |
| Montiferru 2021 | 1    | 0.489             | 0.457 | 0.513 | 0.056  |
| Montiferru 2021 | 5    | 0.475             | 0.399 | 0.597 | 0.197  |
| Muğla 2021      | 1    | 0.514             | 0.410 | 0.608 | 0.198  |
| Muğla 2021      | 6    | 0.571             | 0.545 | 0.597 | 0.052  |
| Muğla 2021      | 8    | 0.623             | 0.533 | 0.724 | 0.192  |
| Muğla 2021      | 10   | 0.634             | 0.433 | 0.716 | 0.283  |

248 Over all 36 combinations the mean is 0.556, the range is 0.374 to 0.724, and ten combinations  
 249 are below chance. The mean spread across the four sources for one scar is 0.183 over nine  
 250 scars and 0.187 over seven. Row D is averaged over sources so that it can be compared with

row C, which is fitted on one region. This does not mean that the choice of foreign source is unimportant.

**Table S5. Prevalence is not the cause of the evaluation-area effect.** The same fitted model and the same out-of-fold predictions are scored in three ways: on the whole region, on a random sample of region cells with the burned fraction of the scar area, and on the scar area. Twenty draws per scar, seed 42. Nine scars are used, because this control needs no leave-one-scar-out model. Values at 4 dp, as stored in the source paper/labelfix\_rerun/code/prevalence\_control.json; checked row by row by paper/code/appendix\_tables.py.

| Target region   | Scar | Burned fraction | A whole region | A' prevalence-matched | B scar area   |
|-----------------|------|-----------------|----------------|-----------------------|---------------|
| Manavgat 2021   | 1    | 0.74            | 0.8822         | 0.8837                | 0.7434        |
| Bejís 2022      | 1    | 0.66            | 0.8245         | 0.8251                | 0.5748        |
| Muğla 2021      | 6    | 0.72            | 0.7773         | 0.7782                | 0.5640        |
| Muğla 2021      | 1    | 0.59            | 0.7773         | 0.7787                | 0.6540        |
| Muğla 2021      | 10   | 0.68            | 0.7773         | 0.7835                | 0.6735        |
| Muğla 2021      | 8    | 0.57            | 0.7773         | 0.7796                | 0.7609        |
| North Evia 2021 | 1    | 0.87            | 0.8642         | 0.8632                | 0.7451        |
| Montiferru 2021 | 1    | 0.58            | 0.7199         | 0.7179                | 0.6215        |
| Montiferru 2021 | 5    | 0.34            | 0.7199         | 0.7285                | 0.4611        |
| <b>Mean</b>     |      |                 | <b>0.7911</b>  | <b>0.7932</b>         | <b>0.6443</b> |

A minus A', the effect of prevalence alone, is  $-0.002$   $[-0.005, +0.001]$ . A' minus B, which replaces both groups of cells, is  $+0.149$   $[+0.087, +0.211]$ . Section S1.9.1 shows that it comes from the unburned cells.

### S1.10 The transfer-gap decomposition, in full

Among the directions that started below chance, the recovered fraction of Section 3.10 is at most  $+0.28$  (Bejís to Evia). Seven directions have negative recovery, five of them with intervals fully below zero. Manavgat to Muğla ( $-0.03$   $[-0.06, +0.01]$ ) and Muğla to Bejís ( $-0.07$   $[-0.16, +0.01]$ ) are negative at the point estimate only. Values are from four\_aoi\_decomposition.csv of the re-frozen outputs

(paper/labelfix\_rerun/round5/tables/corrected/). The within-region reference in the denominator is not matched to a transfer evaluation (Section 4.3), and the raw column depends on the study area (Section 4.4). The fractions are therefore only valid within this protocol.

**Table S6. Transfer-gap decomposition (four-AOI set, 12 directions).** Within = within-region thermal AUC of the target. Best adapted = the better of z-score and CORAL; this choice uses target labels. Recovered fraction = (adapted – raw)/(within – raw), signed and not clipped, with a paired bootstrap CI (1000 replicates, 2-cell blocks). Montiferru directions are not part of this decomposition. The status column shows whether the *adapted* value is above chance, using the 2-cell adapted intervals of Table S16.

| Direction      | Within | Raw   | Best adapted (method) | Recovered fraction [CI] | Status                       |
|----------------|--------|-------|-----------------------|-------------------------|------------------------------|
| Bejís→Evia     | 0.912  | 0.383 | 0.532 (z-score)       | +0.28 [+0.23, +0.32]    | adapted above chance         |
| Muğla→Manavgat | 0.908  | 0.345 | 0.485 (z-score)       | +0.25 [+0.22, +0.28]    | adapted, chance not excluded |
| Evia→Bejís     | 0.918  | 0.448 | 0.549 (z-score)       | +0.22 [+0.16, +0.27]    | adapted above chance         |
| Bejís→Manavgat | 0.908  | 0.314 | 0.406 (CORAL)         | +0.15 [+0.13, +0.18]    | adapted still below chance   |
| Manavgat→Bejís | 0.918  | 0.396 | 0.467 (CORAL)         | +0.14 [+0.09, +0.17]    | adapted still below chance   |
| Manavgat→Muğla | 0.859  | 0.438 | 0.427 (z-score)       | –0.03 [–0.06, +0.01]    | <b>negative recovery</b>     |
| Muğla→Bejís    | 0.918  | 0.583 | 0.560 (CORAL)         | –0.07 [–0.16, +0.01]    | <b>negative recovery</b>     |
| Evia→Muğla     | 0.859  | 0.577 | 0.530 (CORAL)         | –0.17 [–0.22, –0.11]    | <b>negative recovery</b>     |
| Muğla→Evia     | 0.912  | 0.653 | 0.563 (CORAL)         | –0.35 [–0.43, –0.27]    | <b>negative recovery</b>     |
| Bejís→Muğla    | 0.859  | 0.618 | 0.518 (z-score)       | –0.42 [–0.51, –0.34]    | <b>negative recovery</b>     |
| Manavgat→Evia  | 0.912  | 0.654 | 0.529 (z-score)       | –0.48 [–0.56, –0.42]    | <b>negative recovery</b>     |
| Evia→Manavgat  | 0.908  | 0.677 | 0.417 (CORAL)         | –1.13 [–1.27, –1.00]    | <b>negative recovery</b>     |

### S1.11 The thermal sign, stratified

Section 4.4 reports that four of five regions show a negative association between pre-fire surface temperature and burning, and that Manavgat shows the same once elevation is held. The values per region are given here. Signed AUC against burned within the 10 km collar. The stratified columns pool the concordance within deciles of the named variable, and each decile is weighted by its number of burned and unburned cell pairs. Sources:

paper/labelfix\_rerun/code/matched\_frame\_gap.csv (LST raw, within NDVI, within distance; NDVI columns), which can be recomputed by paper/code/verify\_matched\_gap.py, and paper/labelfix\_rerun/round7/r7b\_lst\_given\_terrain.csv (LST within elevation, and LST detrended on elevation).

**Table S27. The LST association, stratified, 10 km collar.** Point estimates; 10-cell block-bootstrap intervals for every cell are in paper/labelfix\_rerun/round8/r8e\_stratified\_intervals.csv, and the Manavgat values are summarised in Table S35.

| Region     | LST raw | LST within elevation | LST detrended on elevation | LST within NDVI | LST within distance | NDVI raw | NDVI within LST |
|------------|---------|----------------------|----------------------------|-----------------|---------------------|----------|-----------------|
| Manavgat   | 0.522   | <b>0.403</b>         | <b>0.409</b>               | 0.556           | 0.454               | 0.551    | 0.550           |
| Bejís      | 0.405   | 0.509                | 0.411                      | 0.486           | 0.350               | 0.618    | 0.574           |
| Muğla      | 0.332   | 0.363                | 0.365                      | 0.404           | 0.367               | 0.652    | 0.547           |
| Evia       | 0.286   | 0.327                | 0.312                      | 0.279           | 0.319               | 0.663    | <b>0.380</b>    |
| Montiferru | 0.376   | 0.392                | 0.416                      | 0.368           | 0.485               | 0.582    | <b>0.405</b>    |

Three results follow. First, the raw LST association of Manavgat is above 0.5, and it falls well below 0.5 once elevation is held. This holds both for deciles and for removing the linear dependence of LST on elevation in the region ( $-5.6$  K per km). On the full study area the same holds: 0.665 raw, against 0.455 and 0.446. The positive signal in Manavgat therefore comes from terrain. With elevation held, Manavgat has the same sign as the other four regions at the point estimates; the interval excludes 0.5 on the collar but not on the original study area (Table S35). Second, in the other four regions the sign stays below 0.5 within elevation deciles in three regions and within NDVI deciles in all four. In Bejís it moves to 0.509 within elevation deciles and stays below 0.5 after detrending. Third, NDVI reverses in two regions once LST is held, but LST reverses in none of the four once NDVI is held. Holding northness

(the cosine of aspect) through deciles does not change the sign of the LST association in any region (Table S35).

## S1.12 The LST anomaly under the difference instrument

Section 4.4 reports two elevation reversals on the collar, both involving Manavgat, that meet the per-comparison criterion. The LST anomaly differs between regions under a weaker test: the two regions are bootstrapped separately and their difference is taken. The three pairs below have point estimates on opposite sides of 0.5 and a difference interval that excludes zero, all for `1st_anomaly_mean`. Signed AUC within the 10 km collar; the two regions are bootstrapped separately with 10-cell spatial blocks, 1000 replicates, seed 42.

**Table S7. Between-region differences in the signed LST-anomaly association, 10 km collar.** Pairs with point estimates on opposite sides of 0.5 and a difference interval that excludes zero. Source `paper/labelfix_rerun/inference/reversal_family_holm.csv` (collar frame); checked row by row by `paper/code/appendix_tables.py`.

| Pair               | AUC A | AUC B | Difference | 95 % CI          |
|--------------------|-------|-------|------------|------------------|
| Bejís vs Evia      | 0.392 | 0.584 | -0.191     | [-0.295, -0.083] |
| Evia vs Montiferru | 0.584 | 0.400 | +0.184     | [+0.012, +0.321] |
| Bejís vs Muğla     | 0.392 | 0.507 | -0.115     | [-0.219, -0.002] |

On the collar, 27 of the 90 feature-by-pair differences have intervals that exclude zero, against about 4.5 expected by chance at 5 %. Four of the 27 are for this feature. Three of these four have point estimates on opposite sides of 0.5, and two of the three involve Evia. With a Holm correction on normal-approximation p-values, 12 of the 90 remain. With the stricter intersection-union form, which also requires the association in each region to be significant, none remains (Section S1.19). The nine features carry about two independent thermal signals (Section 3.4), so the 90 comparisons are not independent. This result is weaker than a reversal.

### S1.13 The same-geography event pair

Section 4.4 gives the result and shows why it does not survive the frame test. The design and two differences from the twenty-direction matrix are given here. The protocol is in Section S3.3.

This comparison was designed to keep the place fixed and change only the fire. Muğla burned in 2021 and again eleven months later, on the same grid and with the same processing. The 2022 population is the 2021 population without the 2021 scar: 41,730 rows with 2,911 burned for 2021, against 38,790 rows with 331 burned for 2022. **The 2021 population has 70 blocks of 5 km with burned cells, and the 2022 population has 11.** Eleven is below the minimum of sixteen used in this design, so the 2022 intervals are indicative, like the 20-cell row of Table 1. Values for all features are in Table S12.

**On the original study area, elevation reverses with bootstrap support.** In 2021, higher ground burned more, at 0.611 [0.532, 0.690]. In 2022, lower ground burned more, at 0.296 [0.230, 0.355]. The intervals do not overlap, and the difference is  $-0.317$  [ $-0.414$ ,  $-0.220$ ].

**The collar removes this reversal.** The 2022 fire is one compact scar inside the whole Muğla rectangle, so 93.2 % of its cells lie more than 10 km from any burned cell (median 43.6 km), against 55.3 % in 2021. This is the largest share of distant cells in the cohort. On the 10 km collar, the 2021 value hardly changes (0.611 to 0.606, still supported). The 2022 value moves from 0.297 to 0.565, to the same side of 0.5 as in 2021, with an interval that includes chance. Slope does not reverse on either frame.

**Only elevation and slope can be evaluated.** The modelling export of the 2022 fire is not in this repository. The released script (`paper/code/verify_mugla_collar.py`) rebuilds the 2022 population from the 2021 predictor file with the 2022 burned mask. This is valid for elevation, slope and land cover, which are the same in both years, but not for the seasonal channels.

Compared with the pipeline's own 2022 export, the rebuilt data agree on elevation (0.297 against 0.296) and slope (0.559 against 0.558). They differ on every seasonal channel: by +0.136 on 1st\_anomaly\_mean, +0.092 on tvdi\_difference\_mean, -0.078 on ndvi\_mean and -0.025 on current\_tvdi\_mean. **Only elevation and slope are therefore given a collar result here.**

| Feature                      | 2021, full              | 2022, full              | full verdict              | 2021, collar            | 2022, collar            | collar verdict            |
|------------------------------|-------------------------|-------------------------|---------------------------|-------------------------|-------------------------|---------------------------|
| elevation_mean               | 0.611<br>[0.529, 0.692] | 0.297<br>[0.229, 0.363] | <b>supported reversal</b> | 0.606<br>[0.525, 0.685] | 0.565<br>[0.450, 0.677] | none                      |
| slope_mean                   | 0.637<br>[0.584, 0.689] | 0.559<br>[0.463, 0.643] | none                      | 0.635<br>[0.578, 0.692] | 0.457<br>[0.356, 0.548] | none                      |
| the state-dependent channels | n/a                     | n/a                     | n/a                       | n/a                     | n/a                     | <b>not evaluable here</b> |

Source paper/labelfix\_rerun/code/mugla\_two\_event\_collar.csv. The two populations are not separate samples. They share 38,789 of the 38,790 cells of the 2022 population. Elevation, slope and land cover are identical in all 73,098 grid cells, so only NDVI and the six thermal channels differ between them. The 2021 burned cells were also removed from the 2022 population. In the 2022-to-2021 direction, therefore, no burned target cell is in the source training data, while 38,789 of the 38,819 unburned target cells are. Membership of the training data alone separates the two classes of the 2021 target at ROC-AUC 0.9996. **The effect of this on a transfer estimate between the two years cannot be bounded.** The known part of the bias makes the reversal smaller, not larger: the removed cells are high (median 563 m), so removing them takes high unburned cells out of the 2022 population.

#### **S1.14 The two interventions, in full**

**(a) Pooled multi-region training** (Fig. 6). A model trained on the pooled populations of the other four regions is better than the mean of the four single-source models for one target of five, Evia, at 0.715 [0.668, 0.757] against a mean of 0.569. For the other four targets it is

worse than this mean, by 0.009 to 0.057. It is better than the best single source only for Evia (0.654). The best source can only be chosen with target labels, so it is a reference, not an option. For every target the pooled model stays 0.20 to 0.48 AUC below the within-region ceiling. For Manavgat and Bejís it is below chance, and only for Manavgat with interval support, at 0.426 [0.369, 0.486]. Outside Evia, pooling does not recover what single-source transfer loses (paper/labelfix\_rerun/code/loro\_all.json, paper/labelfix\_rerun/round7/r7d\_pooled\_vs\_single.csv).

**(b) Removing the direction-reversing features.** Under the original label, two predictors reversed between regions with bootstrap support on the original study areas: **elevation\_mean** and **1st\_anomaly\_mean**. This analysis keeps these two and does not re-select under the corrected label (Table S10). Section 4.4 shows that the study area decides much of the support, so this analysis measures the cost of removal under the original protocol. Without the two predictors, mean within-region AUC falls by **0.076**, with support in every region (−0.035, −0.130, −0.073, −0.063 and −0.079 for Manavgat, Bejís, Muğla, Evia and Montiferru; every interval fully below zero). Mean transfer changes by **+0.014 [−0.028, +0.056]** (Student *t* over the ten region pairs, as plotted in Fig. 7), which includes zero; the primary pair-cluster bootstrap gives [−0.018, +0.049] (source paper/labelfix\_rerun/round3/feature\_drop\_transfer.json, summarised by paper/labelfix\_rerun/round7/r7e\_supplement\_tables.py).

Two points should be noted. First, most of the cost does not come from the thermal set. Removing elevation alone lowers AUC by 0.056 (0.896 to 0.840). Removing the LST anomaly alone lowers it by 0.013 (0.896 to 0.883). About three quarters of the cost therefore comes from removing a baseline terrain variable. Second, both predictors were chosen because they reverse, on the same data on which the costs are estimated, and no correction is applied.

**Table S28. Feature-removal configurations.**

| Configuration         | Mean within-region AUC | Mean transfer AUC |
|-----------------------|------------------------|-------------------|
| full                  | 0.896                  | 0.527             |
| drop elevation_mean   | 0.840                  | 0.533             |
| drop lst_anomaly_mean | 0.883                  | 0.529             |
| drop both             | 0.820                  | 0.541             |

Fig. 7 shows the four configurations together. A local cost is measured, but no transfer gain. This is not an exchange, because neither transfer effect is established: the thermal contribution to transfer is +0.007 and removing the reversing predictors gives +0.014, and both intervals include zero.

### **S1.15 The paired thermal contribution to transfer**

The signed associations on both frames are given in Tables S9, S10 and S14, and their stratified values in Section S1.11; Section 4.4 summarises them. This section gives the uncertainty of the paired thermal contribution to transfer.

**The paired thermal contribution to transfer.** The thermal and baseline matrices were differenced direction by direction. The mean is **+0.007**. The single contributions range from **−0.148 to +0.132**, with **twelve positive and eight negative**. A mean near zero therefore comes from positive and negative values that cancel, not from a consistent absence of effect. The directions are not independent, because each region appears in eight of the twenty. The interval therefore depends on the resampling unit. All four units that the design allows give intervals that include zero (1000 replicates where resampled; source paper/labelfix\_rerun/code/transfer\_delta\_ci.json):

**Table S29. The mean paired thermal contribution to transfer, by resampling unit.**

| Resampling unit                                 | n  | 95 % interval on the mean paired contribution |
|-------------------------------------------------|----|-----------------------------------------------|
| Directions, naive bootstrap                     | 20 | [−0.023, +0.037]                              |
| Unordered pairs, cluster bootstrap (primary)    | 10 | [−0.020, +0.038]                              |
| Unordered pairs, Student <i>t</i> on pair means | 10 | [−0.028, +0.043]                              |

| Resampling unit                  | n | 95 % interval on the mean paired contribution |
|----------------------------------|---|-----------------------------------------------|
| Regions, leave-one-out jackknife | 5 | [-0.018, +0.033]                              |

Two further units can be computed on this frame (20,000 replicates; paper/labelfix\_rerun/round8/r8d\_summary.json): clustering by target region gives [-0.006, +0.022], clustering by source region [-0.008, +0.026] and the pigeonhole bootstrap [-0.028, +0.045]. All six computable units therefore include zero.

Leaving out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0050, +0.0072, +0.0010 and +0.0087. No single region therefore carries the mean or changes its sign. These units resample only between directions. They do not include the sampling variability within a direction.

### **S1.16 The contrast pair, in full**

Manavgat and Muğla are in the same country, burned in the same year, and are 306 km apart. Their burned cells have the most similar environmental range of any pair in the matrix. Bejís and Montiferru have the least similar range (Fig. 8). Values for both pairs, on both frames, are in Table S18.

For Manavgat and Muğla on the original study areas, seven of nine feature-response directions point opposite ways, and all six features supported in both regions have opposite signs, including elevation. On the collar the elevation reversal remains, at 0.376 [0.300, 0.465] in Manavgat against 0.606 [0.525, 0.685] in Muğla. Transfer is below chance in both directions on both frames: 0.438 and 0.345 as drawn, 0.493 and 0.433 under the collar. On the collar, the interval of Muğla to Manavgat, [0.381, 0.491], excludes chance. The most similar pair is among the weakest in the matrix, but it is not the weakest. On the collar, the weakest direction is Manavgat to Bejís at 0.407 and the strongest is Muğla to Evia at 0.728. The niche-overlap and applicability measures of Table S18 are computed on the original study areas.

Bejís and Montiferru are at the other extreme. Their burned ranges hardly overlap, and they have the most different values on every overlap measure. They transfer above chance in both directions, but only at the point estimate: neither direction has interval support at 5 km blocking. The applicability measure was not produced for Montiferru. **At the point estimates, high niche overlap did not ensure transfer, and low overlap did not prevent it.** This rests on two pairs. It is not a correlation across pairs; that is tested in Section S1.21.

### **S1.17 The sensitivity arms, summarised**

Eight design choices were changed, one at a time, with everything else kept fixed: the Evia study area and its prevalence, the CORAL regularisation, the block size, the closing date of the predictor window, the quality screening of the coarse thermal input, normalised against absolute dryness channels, the removal of the coordinate-informed channels, and the capacity of the classifier (Sections S1.1 to S1.8). None changes a conclusion of the main text. Two of them affect how the results should be read.

**The secondary population.** Section 3.5 defines a secondary population of all valid cells, including cropland. It exists only for the within-region models in Manavgat and Bejís, so it is a two-region sensitivity analysis, and no transfer quantity is defined on it. Across the two regions and three block sizes, the thermal gain is +0.048 to +0.061, with every bootstrap interval above zero, against +0.045 to +0.067 in the primary population. The paired difference between the populations is −0.011 to +0.011, with no consistent sign (paper/labelfix\_rerun/pipeline/robustness/step8\_large\_block\_primary\_all\_valid/; Table 1). The gain is therefore not caused by excluding cropland. This analysis says nothing about transfer.

## S1.18 Distance within a region

This analysis measures how skill falls with distance inside one region. A model is fitted on one half of a region and applied to the other half. Target cells are grouped by their distance from the training cells (paper/code/distance\_curve.py; corrected output paper/labelfix\_rerun/round3/distance\_curve.json, summarised in paper/labelfix\_rerun/round7/r7h\_distance\_curve.csv). Means are unweighted over bins. The bins contain 1 to 1,067 burned cells and have no intervals.

**Table S32. Within-region skill by separation from the training cells.**

| Separation from training cells | Bins | Mean target AUC |
|--------------------------------|------|-----------------|
| 0 to 5 km                      | 18   | 0.709           |
| 5 to 10 km                     | 16   | 0.570           |
| 10 to 20 km                    | 11   | 0.519           |
| 20 to 40 km                    | 6    | 0.499           |
| 40 to 80 km                    | 3    | 0.541           |
| 80 to 160 km                   | 1    | 0.421           |
| cross-region, 306 to 2,802 km  | 20   | 0.527           |

**At 10 to 20 km inside one region, the model is close to chance.** This limits how far such a model can be used from its training cells. It agrees with Section 4.3, where withholding a scar or using a foreign model had no measurable cost.

This does not show that the cross-region failure is only an effect of distance. Once the curve reaches chance, any cross-region mean near 0.5 must lie on its continuation, so this comparison cannot test it. Also, a model that carries no information does not give a reliably reversed ranking, but Manavgat to Bejís is below chance with interval support both on the original study area and on the 10 km collar.

Four limits apply. First, the two distance ranges do not overlap: within-region distances are 2 to 86 km, and cross-region distances start at 306 km. Second, the far bins contain few cells, so their means should not be read closely; the rise at 40 to 80 km is not evidence of an effect.

Third, the near bins include the spatial autocorrelation that blocked validation is designed to remove, so **0.709 is an upper bound on near-field skill, not an estimate of it**. Fourth, the design separates distance from crossing a study-area boundary, but not from the land cover, terrain and fire history that change with distance. The analysis gives a length scale for the fall of skill within a region, not a cause.

### **S1.19 The frame test, elaborated**

Section 4.4 gives these results. Sources: `paper/labelfix_rerun/code/aoi_frame_auc.csv` (signed associations by frame), `paper/labelfix_rerun/code/collar_frame_bootstrap.csv` (their 10-cell intervals), `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv` (the transfer matrix by frame) and `paper/labelfix_rerun/code/matched_frame_gap.csv` (the matched reference); code in `paper/code/`.

**Construction.** Every region is restricted to cells within 10 km of any burned cell. This removes only distant unburned cells. Every burned cell is at distance zero and is kept at any radius, so radii were compared to guard against a favourable choice. The interval criterion was evaluated at the 10 km collar; the 5 and 10 km radii agree at the point estimate. The collar strongly reduces the number of cells in two regions, so part of any loss of support is a loss of power. On average, 2.3 features per direction are supported in both regions on the original study areas, and 2.2 on the collar.

**Multiplicity.** Each frame has a family of ninety feature-by-pair comparisons (nine features, ten region pairs). Table S30 gives how many pass each criterion. The per-comparison criterion of Section 3.10 leaves two supported reversals on the collar, both on elevation and both involving Manavgat: Manavgat against Muğla, which is also supported on the original areas, and Manavgat against Evia, which is supported only on the collar. Manavgat against Bejís, supported on the original areas, misses support on the collar by 0.004. Neither survives

the Holm intersection-union correction, which adjusts the tests of both regions together (adjusted  $p = 0.33$  for Manavgat against Evia and  $0.79$  for Manavgat against Muğla, normal approximation).

**Table S30. Reversal family, ninety feature-by-pair comparisons per frame.** Source paper/labelfix\_rerun/inference/reversal\_family\_holm.csv, reversal\_per\_region.csv.

| Criterion                                                                          | Full frame | 10 km collar |
|------------------------------------------------------------------------------------|------------|--------------|
| Difference interval excludes zero                                                  | 39         | 27           |
| ... with opposite-sided point estimates                                            | 31         | 21           |
| Per-comparison criterion (each region's own interval excludes 0.5, opposite sides) | 14         | 2            |
| Holm, normal approximation, difference test                                        | 27         | 12           |
| Holm intersection-union (both regions' tests)                                      | 5          | 0            |

**Similarity measures on the collar.** Two of the twenty similarity measures are built from the signed associations, and they were recomputed on the collar. The sign-agreement fraction over supported features is  $\rho = +0.52$   $[-0.27, +0.87]$  on the original study areas. On the collar it takes only the values 0 and 1 across sixteen defined directions, and it correlates with collar transfer at  $\rho = +0.38$   $[-0.17, +0.85]$ . The all-feature cosine is  $+0.44$   $[-0.29, +0.80]$  on the original study areas and correlates with collar transfer at  **$+0.57$   $[+0.05, +0.88]$** . This is the only interval of a similarity measure in the study that excludes zero (paper/labelfix\_rerun/round6/diag\_collar/; intervals from paper/labelfix\_rerun/round7/r7g\_collar\_diagnostics.json, which reproduces the original intervals exactly). It is not treated as a result for three reasons. It is one of about forty uncorrected tests. It rests on twenty directions that share regions. The collar is defined from the burned cells, so the measure needs target labels twice. The other eighteen measures were not recomputed on the collar.

**Data provenance.** A quality-screening rebuild overwrote the 500 m modelling datasets of Manavgat and Muğla at the standard pipeline path after the frozen tables were computed. Each new file differs in `downscaled_1st_mean` and `fused_1st_mean`. The pipeline records a SHA-256 for the modelling dataset of each region, and this hash identifies the frozen copy. Every analysis of Section 4.4 reads each region through `paper/code/_canonical.py`, which checks the hash on load, and reads Manavgat from the corrected re-freeze (SHA-256 5a5e876c...). The provenance of the collar transfer matrix is in `paper/labelfix_rerun/round5/collar/PROVENANCE.md`.

**Table S8. Cross-region transfer on comparable study areas.** Natural-vegetation population, thermal model, twenty ordered directions per row. Above and below chance are point counts. The supported counts use a 10-cell ( $\approx 5$  km) spatial-block bootstrap on the target, 1000 replicates, seed 42. Table S16 gives the original matrix under 2-cell ( $\approx 1$  km) blocking. Bounds per direction are in `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv`.

| Source frame | Target frame | Mean target AUC | Above chance    | Below chance | Supported above / below | Paired thermal delta |
|--------------|--------------|-----------------|-----------------|--------------|-------------------------|----------------------|
| full         | full         | 0.527           | 13 of 20        | <b>7</b>     | <b>9 / 6</b>            | +0.007               |
| full         | 10 km        | 0.559           | 14 of 20        | 6            | 10 / 2                  | +0.008               |
| 10 km        | full         | 0.546           | 14 of 20        | 6            | 10 / 5                  | +0.014               |
| <b>10 km</b> | <b>10 km</b> | <b>0.589</b>    | <b>15 of 20</b> | <b>5</b>     | <b>13 / 2</b>           | <b>+0.024</b>        |
| 5 km         | 5 km         | 0.591           | 16 of 20        | 4            | 11 / 0                  | +0.017               |

The baseline changes with the frame. On the 10 km collar, the static baseline transfers at 0.565 and the thermal model at 0.589, a paired difference of +0.024 instead of +0.007. The static baseline still does not transfer well. The number of directions below chance falls from seven to five at the point estimate, and from six to two with interval support. The largest rises are Bejís to Evia (0.383 to 0.602) and Bejís to Manavgat (0.314 to 0.452). The only large fall is Evia to Manavgat (0.677 to 0.594).

**The matched reference.** The within-region reference was recomputed on the same frame at 5 km blocking (paper/code/verify\_matched\_gap.py):

**Table S33. Within-region reference and transfer on matched frames.**

| Frame          | Within-region (5 km blocking) | Mean transfer | Gap          |
|----------------|-------------------------------|---------------|--------------|
| full rectangle | 0.814                         | 0.527         | 0.287        |
| 10 km collar   | <b>0.786</b>                  | <b>0.589</b>  | <b>0.197</b> |
| 5 km collar    | 0.759                         | 0.591         | 0.169        |

Paired by target region, the shortfall on the collar is **+0.197 [+0.091, +0.303]** (Student *t* over the five target regions; +0.081 Montiferru, +0.319 Manavgat, +0.172 Muğla, +0.202 Bejís, +0.211 Evia). The shortfall remains on every matched row. It is 0.197 on the collar, not the 0.29 of the unmatched comparison, and it becomes smaller as the frame moves closer to the fire. On the 5 km collar, the within-region gain falls by half but stays positive in all five regions, with a mean of +0.042 (Manavgat +0.035).

## **S1.20 The transfer matrix and adaptation, elaborated**

Section 4.5 gives these results.

**Raw transfer.** Raw target AUC ranges from 0.314 to 0.677 (Fig. 4). At 2-cell blocking, eleven of 20 directions are above chance with interval support and seven are below: both directions between Manavgat and Bejís, both between Manavgat and Muğla, both between Bejís and Evia, and Montiferru to Manavgat. At 10-cell blocking the same point estimates give 9 above, 6 below and 5 uncertain. Manavgat to Muğla loses its support, and no direction changes side of the chance line. Of the directions below chance, Manavgat to Bejís (0.407 [0.323, 0.493]) and Muğla to Manavgat (0.433 [0.381, 0.491]) keep interval support on the 10 km collar, but both intervals include chance on the 5 km collar. The lowest direction on the original study areas is Bejís to Manavgat, at 0.314 [0.296, 0.332].

**Label-free adaptation.** Under region-wise z-scoring the twenty directions range from 0.302 to 0.630, and under CORAL from 0.406 to 0.624. The change is not uniform: under z-scoring, Bejís to Manavgat falls further below chance, from 0.314 to 0.302. With the better of the two methods in each direction, 15 of the 20 directions end closer to chance and 5 end further from it. Four of these five involve Montiferru and move up. The fifth, Manavgat to Muğla, moves down from 0.438 to 0.427. Under CORAL alone, sixteen of twenty end closer to chance (paper/labelfix\_rerun/round5/matrix20\_official.csv). In the six directions where raw transfer was below chance, the best label-free method recovers at most 28 % of the gap to the within-region reference (Bejís to Evia). On the twelve directions of the decomposition, seven show negative recovery (Section S1.10).

**Seed stability.** Across five bootstrap seeds, every 2-cell verdict is stable. At 10 cells, one level verdict and two paired-delta verdicts change: the level verdict of Evia to Bejís, and the paired-delta verdicts of Manavgat to Muğla and Evia to Montiferru (paper/labelfix\_rerun/round6/seed\_stability.json). The random-forest seed is fixed, so all intervals per direction are conditional on one fitted source model.

**Jackknife.** Leaving out one region at a time moves the mean paired contribution between +0.001 (without Evia) and +0.015 (without Manavgat). Its sign never changes.

## **S1.21 The similarity diagnostics**

Twenty similarity measures from five families were each rank-correlated with raw thermal transfer AUC over the directions on which they are defined. One pair-based bootstrap is used for all of them (unordered pairs resampled with replacement, both directions kept, 2000 replicates, seed 42). The measures were defined on 8 August 2026, together with their results under the original label, and were not re-selected after the label correction. Table S17 lists all twenty. The source is

paper/labelfix\_rerun/round5/out\_official/all\_diagnostics\_vs\_transfer.csv, and the original-label values are in paper/labelfix\_rerun/round5/s6\_diagnostics\_20.md.

**On the original study areas, no interpretable measure has an interval that excludes zero.**

The largest correlations are for the conditional family: the sign-agreement fraction over supported features, at  $\rho = +0.52$   $[-0.27, +0.87]$ , and its cosine, at  $+0.49$   $[-0.24, +0.87]$ . Under the original label these two were the only rows with intervals that excluded zero ( $+0.84$  and  $+0.81$ ). This did not hold after the label correction. The marginal family (applicability, dissimilarity, climatic distance, domain classification), the niche-overlap family and the regime family all include zero. Geographic distance does not order the matrix ( $\rho = -0.17$   $[-0.87, +0.84]$  on the twelve directions where it is defined). The domain classifier separates source from target at  $AUC \geq 0.96$  for every pair. Because it always succeeds, it gives no ordering. The twentieth measure, the vector Spearman of signed AUCs over supported features, is defined on only six directions. Its interval is degenerate (the upper bound equals the point estimate), so it is listed but not interpreted.

**The collar.** Section S1.19 gives the two measures recomputed on the collar. One of them, the all-feature cosine, orders collar transfer with an interval that excludes zero. It is reported with its limits and is not used.

**Limits.** Signed associations need burned labels in both regions, so the conditional family cannot be used before a fire. The marginal family can be used, but it fails. Ten independent region pairs give little power, and a moderate ordering would usually be missed. These results therefore show a failure to find an ordering, not proof that none exists (Section S3.5(xiii)). Measures based on interval support are unstable when bounds lie near 0.5 (Section S3.5(viii)).

603 **S1.22 Additional robustness analyses**

604 The analyses in Table S21 were run after the main results. They test whether the main  
605 quantities depend on analysis choices. None of them was used to select a reported  
606 configuration.

607 **Table S21. Additional robustness analyses (post hoc).** Sources under  
608 paper/labelfix\_rerun/: geometry/ (a1 to a5), inference/ (units, equivalence, ladder) and  
609 round7/.

| Analysis                                           | Question                                                        | Result                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Edge-excluded frame cost (a1)                      | Is the frame cost carried by the scar's edge cells?             | Over nine scars, excluding 0, 1 or 2 cells of the scar edge gives $A - B = 0.147$ [0.086, 0.208], 0.137 [0.069, 0.205] and 0.129 [0.055, 0.204]; with the region as unit 0.160, 0.150 and 0.145, every interval above zero                                                                                 |
| Placebo collars (a2)                               | Is the cost a property of any fire-shaped frame?                | The scar's real positives scored against the negatives of a same-shaped collar placed away from any burned cell reproduce $A$ ( $A - \text{placebo} -0.002$ [-0.056, +0.052], eight scars with placements); the real collar costs +0.155 [+0.108, +0.202] more, so the cost is the fire-adjacent negatives |
| Frame cost against the share of distant cells (a3) | Is the cost larger where the study area has more distant cells? | No. Region-level slope 0.005 [-0.41, +0.42], $r = 0.02$ ( $n = 5$ ); scar-level slope $-0.049$ [-0.34, +0.24] ( $n = 9$ , scars not independent). The test has little power                                                                                                                                |
| Label-free target frames (a4)                      | Does a frame drawn without target labels change transfer?       | Trimming the target to the source's per-feature range, or to the source's area of applicability (Meyer and Pebesma 2021), gives means of 0.520 to 0.574, and a 20 km window 0.482; every paired thermal $\Delta$ interval spans zero                                                                       |
| Metric dependence (a5)                             | Is the frame cost specific to ROC-AUC?                          | Region-wide minus scar-frame over nine scars: partial AUC ( $\text{FPR} \leq 0.1$ ) 0.088 [0.048, 0.128]; average precision $-0.342$ [-0.477, $-0.206$ ], higher on the scar frame because its prevalence is higher                                                                                        |
| Collar within-region increment                     | Does the within-region increment survive the collar?            | +0.083 [+0.037, +0.129] over five regions (Student $t$ ), against +0.087 [+0.037, +0.136] as drawn                                                                                                                                                                                                         |
| Equalised $\Delta$ by resampling unit              | Does the collar $\Delta$ exclude zero?                          | Table S31: two of six computable units exclude zero; the two-way estimators are undefined                                                                                                                                                                                                                  |
| Equivalence of the equalised $\Delta$              | How large a gain is excluded?                                   | Within $\pm 0.05$ under four of six computable units; no gain above 0.047 at 90 % (pair cluster); not within $\pm 0.02$ under any unit                                                                                                                                                                     |
| Within minus transfer increment                    | Is the local-versus-portable contrast itself supported?         | Full frame +0.079 [+0.014, +0.145]; 10 km collar +0.059 [+0.007, +0.110]; 5 km collar +0.026 [-0.012, +0.063]                                                                                                                                                                                              |

| Analysis              | Question                                          | Result                                                                                                                                                              |
|-----------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reversal multiplicity | Do the collar reversals survive correction?       | Table S30: two per-comparison reversals, none after Holm intersection-union                                                                                         |
| scikit-learn version  | Do point estimates depend on the library version? | Under 1.5.2 against 1.9.0, single directions move by up to 0.047 and means by at most 0.003; one support count changes<br>(round7/sklearn152/compare_vs_1_9_0.json) |

610 **Table S31. The equalised (10 km collar) paired contribution by resampling unit.** 20,000  
611 replicates where resampled; 95 % intervals. Source  
612 paper/labelfix\_rerun/inference/units.json.

| Unit                                         | 95 % interval                 | Excludes zero |
|----------------------------------------------|-------------------------------|---------------|
| Directions, naive                            | [-0.001, +0.049]              | no            |
| Unordered pairs, cluster bootstrap (primary) | [-0.003, +0.051]              | no            |
| Target region, cluster bootstrap             | [+0.010, +0.040]              | yes           |
| Source region, cluster bootstrap             | [+0.010, +0.042]              | yes           |
| Pigeonhole (source and target)               | [-0.005, +0.060]              | no            |
| Two-way cluster (CGM)                        | undefined (negative variance) | n/a           |
| Dyadic-robust                                | undefined (negative variance) | n/a           |
| Leave-one-region-out jackknife, $t$          | [-0.017, +0.065]              | no            |

613 The primary unit is the pair cluster (Section 3.7), and on it the paired contribution on the  
614 collar includes zero. With five regions, a percentile bootstrap over regions has at most 126  
615 distinct resamples, so the two region-clustered intervals are coarse.

## 616 **S1.23 Analyses added after the pre-submission review**

617 The analyses in Table S35 were run after an internal review of the manuscript. They test the  
618 main results against the fitted forest, the adaptation method, the inference, the land-cover  
619 map, the burned-area product, terrain aspect and a weather reference. Sources are in  
620 paper/labelfix\_rerun/round8/, which also describes each script.

621     **Table S35. Analyses added after the pre-submission review (post hoc).**

|                                                                                                                                                                    |                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A<br>n<br>a<br>l<br>y<br>s<br>i<br>s                                                                                                                               | Que<br>stion                                                                             | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| R<br>a<br>n<br>d<br>o<br>m<br>-<br>depe<br>f<br>nd<br>o<br>on<br>r<br>the<br>e<br>fitte<br>s<br>d<br>t<br>fore<br>s<br>st?<br>e<br>e<br>d<br>(<br>R<br>8<br>a<br>) | Do<br>the<br>trans<br>fer<br>resul<br>ts<br>depe<br>nd<br>on<br>the<br>fitted<br>forest? | With ten seeds, mean transfer on the original study areas ranged from 0.525 to 0.530 and the mean thermal gain from +0.005 to +0.008. The seed-to-seed standard deviation of a direction had a median of 0.005 (maximum 0.015), and no direction changed side of chance. On the 10 km collar, mean transfer ranged from 0.588 to 0.592, and one direction (Montiferru to Manavgat) changed side. Averaging the predictions of the ten forests gave 0.527 (gain +0.006) and 0.590 (gain +0.025). Seed 42 reproduces the published matrix exactly |

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Question

Result

CORAL with  $\lambda = 1$  gave a mean of 0.512 and CORAL on two thermal principal components 0.523, against 0.527 without adaptation. A placebo that aligns the source with the covariance of a third region instead of the target gave 0.507 and moved 15 of 20 directions closer to chance, against 16 of 20 for CORAL. The movement is therefore not specific to the target. The published CORAL values were reproduced to within 0.009

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 Proceedings of the 8th International Conference on the Analysis of Time Series  
 (Rend. Istit. Lomb. Acc. Sci. Lett. Mat. Ser. A)

Question

Result

No. With all 120 permutations of the five regions, the smallest two-sided p-value was 0.10 (vector Spearman, six directions only); the two supported-feature conditional measures gave 0.125 and 0.15, and all other measures 0.15 or more. The smallest attainable p-value is 1/120 for measures defined on all twenty directions, and 1/24 for those defined on the twelve directions among four regions. Without Manavgat, most correlations are based on 2 to 12 directions; some rise, for example the supported-feature agreement to  $\rho = 0.70$  (10 directions) and climatic distance to  $\rho = 0.72$  (6 directions), but none of these samples can support inference

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Result

Without the eight directions that involve Manavgat, mean transfer was 0.566 on the original study areas (two of twelve directions below chance) and 0.631 on the 10 km collar (none below chance). The thermal gain was +0.015 and +0.032. Transfer is better without Manavgat, but it stays far below within-region skill

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Result

Yes, under all six resampling units that can be computed; the 90 % upper bounds range from 0.021 to 0.037. It is not within  $\pm 0.02$  under any unit

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 1277 (2017) 1–8

| Question                                     | Result                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is the stratified LST association supported? | On the 10 km collar, the Manavgat LST association within elevation deciles is 0.403 [0.343, 0.464], and after removing the linear effect of elevation 0.409 [0.350, 0.475]; both intervals exclude 0.5. On the original study area the values are 0.455 [0.396, 0.509] and 0.446 [0.387, 0.502], which include 0.5 |

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 (R8b, R8g)

| Question                                                                                                                                      | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C<br>a<br>p<br>t<br>u<br>r<br>e<br>o<br>f<br>b<br>u<br>r<br>n<br>e<br>d<br>c<br>e<br>l<br>l<br>s<br>(<br>R<br>8<br>b<br>,<br>R<br>8<br>g<br>) | How many burned cells fall in the high estimated burned cells?<br>In transfer, the 10 % of target cells with the highest thermal scores contained on average 10.3 % of the burned cells (range 0.7 % to 27.1 %), which is what a random ranking gives; eight of twenty directions were below it. The top 20 % contained 19.9 %. Within regions (5 km blocking) the top 10 % contained 35.5 % of the burned cells on average (18.9 % to 44.5 %), and the top 20 % contained 55.7 % |

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Questions and Answers

Result

The 2020 map (v100) was fetched on each region's 30 m grid and aggregated to the same cells; the same procedure applied to v200 reproduces the pipeline's raster and every cell value of the modelling datasets exactly. With the 2020 map, at most 43 burned cells left or entered the natural-vegetation population in Manavgat, Bejís, Muğla and Evia. In Montiferru, 142 burned cells entered it: the share of burned modelled cells with a natural-vegetation fraction of at least 0.5 rose from 0.773 to 0.977, so the 2021 map left out about a fifth of the burned natural vegetation there. This share is computed on the modelled cells and is not the gate fraction of Table S15 (0.723 for Montiferru with the 2021 map), which the gate computes on its own inputs. The within-region thermal gain (5 km blocking) changed from +0.062, +0.045, +0.079, +0.148 and +0.099 to +0.060, +0.071, +0.083, +0.155 and +0.095 (Manavgat, Bejís, Muğla, Evia, Montiferru). Mean transfer changed from 0.527 to 0.513, and the mean thermal gain in transfer from +0.007 to +0.013; seven directions were below chance with both maps, and no direction changed side. Part of the class changes reflects the different algorithms of v100 and v200, not the fires

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| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Result |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Does the MCD64A1 labels fetched again from Earth Engine reproduce the modelling datasets exactly in all five regions. VIIRS VNP64A1 v002 labels, built the same way, agree with them with a Jaccard index of 0.941, 0.950, 0.925, 0.978 and 0.865 (Manavgat, Bejís, Muğla, Evia, Montiferru). With VNP64A1 labels, the within-region thermal gain (5 km blocking) is +0.057, +0.055, +0.089, +0.147 and +0.082; the elevation association of Manavgat is 0.228 (0.232 with MCD64A1); mean transfer is 0.525 (0.527), the mean thermal gain in transfer +0.005 (+0.007), seven directions are below chance with both products, and no direction changes side |        |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>T<br/>e<br/>r<br/>a<br/>i<br/>n<br/>a<br/>s<br/>p<br/>e<br/>c<br/>t<br/>(<br/>R<br/>8<br/>k<br/>,<br/>R<br/>8<br/>m<br/>,<br/>R<br/>8<br/>n<br/>)</p>                                                                                      | <p>Doe<br/>s<br/>aspe<br/>r<br/>ct<br/>expl<br/>ain<br/>the<br/>ther<br/>s<br/>mal<br/>p<br/>gain<br/>or<br/>the<br/>LST<br/>sign<br/>?</p> | <p>Northness and eastness from the Copernicus DEM were added to both feature sets. The within-region thermal gain (5 km blocking) became +0.051 [+0.030, +0.070], +0.044 [+0.025, +0.065], +0.073 [+0.047, +0.099] and +0.139 [+0.109, +0.173] in Manavgat, Bejís, Muğla and Evia, and +0.020 [-0.025, +0.066] in Montiferru, against +0.099 [+0.017, +0.186] without aspect there. Within northness deciles, the LST association is 0.652, 0.482, 0.317, 0.376 and 0.417 (raw 0.665, 0.478, 0.324, 0.377 and 0.370, with missing values filled by the median, so they differ from Table S9 in the third decimal), so aspect does not change its sign</p> |
| <p>W<br/>e<br/>a<br/>t<br/>h<br/>e<br/>r<br/>f<br/>o<br/>l<br/>l<br/>o<br/>w<br/>e<br/>r<br/>w<br/>e<br/>a<br/>t<br/>h<br/>e<br/>r<br/>a<br/>n<br/>o<br/>m<br/>a<br/>l<br/>i<br/>e<br/>s?<br/>n<br/>c<br/>e<br/>(<br/>R<br/>8<br/>o<br/>)</p> | <p>Do<br/>the<br/>ther<br/>mal<br/>chan<br/>nels<br/>follo<br/>w<br/>weat<br/>her<br/>ano<br/>mali<br/>es?</p>                              | <p>Descriptively, across the five regions the mean Landsat LST anomaly follows the ERA5-Land anomaly of air temperature in the predictor window (Spearman <math>\rho = 0.8</math>) but not that of relative humidity (<math>\rho = 0.0</math>), and the mean TVDI difference does not follow humidity either (<math>\rho = 0.0</math>). With five regions this is descriptive only; it fits the reading that the channels describe temperature and surface state more than fuel moisture</p>                                                                                                                                                              |

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| Question                                                                                   | Result                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How many modelled cells lack a thermal value, and is this related to burning?              | In the natural-vegetation population, 1.7 %, 6.5 %, 3.2 %, 4.5 % and 0.1 % of cells (Manavgat, Bejís, Muğla, Evia, Montiferru) have at least one missing thermal channel. The ROC-AUC of this missingness flag against burning is 0.493 to 0.512. Among all valid cells the share is larger (up to 57.5 % in Evia, mostly sea and coast)                                                                            |
| Does a model with source and target random intercepts change the thermal gain in transfer? | No. The estimated source and target variances are close to zero, so the fit is singular and reduces to a pooled interval with four degrees of freedom; it is a more conservative interval, not an independent confirmation. The mean gain is +0.007 [−0.035, +0.050] on the original study areas and +0.024 [−0.012, +0.060] on the 10 km collar (REML, t interval with four degrees of freedom); both include zero |

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Question

Result

Revised: Do 5 km blocks remain over the spatial dependence?

Not fully. The correlation of out-of-fold residuals between cell pairs is 0.58 to 0.83 below 1 km, 0.15 to 0.48 at 5 to 10 km (except Montiferru,  $-0.01$ ), and falls below 0.05 at 10 to 20 km in Manavgat, Bejís and Muğla, at 20 to 40 km in Evia and at 5 to 10 km in Montiferru. Intervals at 5 km blocking may therefore be somewhat too narrow

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| Questions                                                            | Results                                                                                                                          |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Does pooling out-of-fold predictions inflate the within-region AU C? | No. The mean of the five fold AUCs is 0.004 to 0.018 higher than the pooled value in every region (thermal model, 5 km blocking) |

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| Question                                                                          | Result                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How uncertain are the four estimates of Table S25?                                | Pair-cluster intervals of mean transfer: 0.527 [0.461, 0.588], 0.527 [0.460, 0.594], 0.522 [0.456, 0.584] and 0.481 [0.419, 0.542]; of the thermal gain in transfer: +0.007 [−0.022, +0.034], −0.011 [−0.040, +0.016], −0.019 [−0.045, +0.007] and −0.021 [−0.057, +0.010]. All gain intervals include zero |
| Do the 48 cells that burned in the predicted indicator window affect the results? | Ten of them are labelled burned. Without them, the within-region thermal gain was +0.050 at 1 km blocking (+0.056 with them) and +0.061 at 5 km (+0.045). Transfer in the eight directions that involve Bejís changed by at most 0.015, and their mean thermal gain from +0.011 to +0.007                   |

622 A leave-one-block-out reference with a 10 km buffer was also tried. Its pooled AUC  
623 combines the predictions of different models, and because the burned cells lie in one or two  
624 blocks, the pooled value was dominated by calibration differences between the models. It is

625 therefore not reported; the distance curve of Section S1.18, which scores a single model,  
 626 answers the same question.

## 627 S2 Supporting tables

628 These tables give the values per region and per direction that the Results quote, so that every  
 629 statement can be checked against the values behind it.

630 **Table S9. Signed univariate AUC of each predictor against burned, by region.** Natural-  
 631 vegetation population; 10-cell (~5 km) spatial-block bootstrap, 1000 replicates, seed 42. The  
 632 AUC is not folded to  $\max(\text{AUC}, 1 - \text{AUC})$ . A value below 0.5 means that lower values rank  
 633 burned cells higher; it shows a direction, not weakness. **Bold** marks a region whose own  
 634 interval excludes 0.5. Source  
 635 paper/labelfix\_rerun/round5/tables/step9g\_multi\_aoi\_feature\_stability.csv (the  
 636 pipeline's Step9G five-region synthesis, sha256 c864cd7d...); checked row by row by  
 637 paper/code/appendix\_tables.py.

| Feature              | Manavgat                    | Bejís                       | Muğla                       | Evia                        | Montiferru                  |
|----------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| elevation_mean       | <b>0.232</b> [0.179, 0.288] | <b>0.643</b> [0.558, 0.729] | <b>0.611</b> [0.532, 0.690] | 0.541 [0.448, 0.626]        | 0.584 [0.395, 0.762]        |
| slope_mean           | <b>0.400</b> [0.340, 0.459] | 0.521 [0.439, 0.605]        | <b>0.637</b> [0.582, 0.686] | 0.487 [0.418, 0.554]        | <b>0.652</b> [0.506, 0.771] |
| ndvi_mean            | 0.564 [0.499, 0.628]        | 0.559 [0.497, 0.619]        | <b>0.662</b> [0.616, 0.704] | <b>0.639</b> [0.575, 0.701] | 0.586 [0.450, 0.704]        |
| lst_anomaly_mean     | 0.509 [0.460, 0.560]        | <b>0.418</b> [0.364, 0.480] | 0.485 [0.395, 0.566]        | <b>0.640</b> [0.567, 0.710] | 0.395 [0.285, 0.535]        |
| current_lst_mean     | <b>0.665</b> [0.608, 0.719] | 0.477 [0.401, 0.547]        | <b>0.325</b> [0.271, 0.382] | <b>0.377</b> [0.301, 0.456] | 0.370 [0.248, 0.513]        |
| current_tvdi_mean    | <b>0.677</b> [0.622, 0.733] | 0.517 [0.429, 0.595]        | <b>0.336</b> [0.275, 0.398] | <b>0.362</b> [0.285, 0.442] | <b>0.356</b> [0.233, 0.499] |
| tvdi_difference_mean | 0.460 [0.409, 0.510]        | 0.512 [0.443, 0.583]        | 0.490 [0.396, 0.575]        | 0.519 [0.444, 0.589]        | <b>0.378</b> [0.282, 0.497] |
| downscaled_lst_mean  | <b>0.683</b> [0.626, 0.739] | 0.484 [0.400, 0.560]        | <b>0.307</b> [0.253, 0.366] | <b>0.377</b> [0.297, 0.459] | 0.365 [0.240, 0.511]        |
| fused_lst_mean       | <b>0.666</b> [0.610, 0.721] | 0.481 [0.404, 0.551]        | <b>0.325</b> [0.272, 0.383] | <b>0.376</b> [0.300, 0.456] | 0.370 [0.248, 0.513]        |

**Definition of a reversal.** A pair of regions is called a reversal only when their signed associations lie on opposite sides of 0.5 **and the interval of each region excludes 0.5**. This is stricter than requiring disjoint intervals. For example, for `current_1st_mean` in Manavgat and Bejís the two intervals are disjoint ([0.608, 0.719] against [0.401, 0.547]), but the Bejís interval includes 0.5. Bejís therefore has no established direction, and this pair is a point reversal only.

**Table S10. Cross-region reversals that meet the strict criterion.** The criterion is applied to the intervals of Table S9. Difference intervals are from the paired 10-cell block bootstrap of `paper/code/ems_inference_multiplicity.py` (`paper/labelfix_rerun/inference/reversal_family_holm.csv`, 1000 replicates), which flags the same fourteen pairs. Checked row by row by `paper/code/appendix_tables.py`.

| Feature                          | Region A   | AUC   | Region B   | AUC   | Difference [95 % CI]    |
|----------------------------------|------------|-------|------------|-------|-------------------------|
| <code>elevation_mean</code>      | Manavgat   | 0.232 | Bejís      | 0.643 | +0.411 [+0.312, +0.509] |
| <code>elevation_mean</code>      | Manavgat   | 0.232 | Muğla      | 0.611 | +0.379 [+0.274, +0.476] |
| <code>slope_mean</code>          | Manavgat   | 0.400 | Muğla      | 0.637 | +0.237 [+0.158, +0.313] |
| <code>slope_mean</code>          | Manavgat   | 0.400 | Montiferru | 0.652 | +0.252 [+0.100, +0.389] |
| <code>lst_anomaly_mean</code>    | Bejís      | 0.418 | Evia       | 0.640 | +0.222 [+0.129, +0.316] |
| <code>current_1st_mean</code>    | Muğla      | 0.325 | Manavgat   | 0.665 | +0.340 [+0.260, +0.413] |
| <code>current_1st_mean</code>    | Evia       | 0.377 | Manavgat   | 0.665 | +0.288 [+0.195, +0.384] |
| <code>current_tvdi_mean</code>   | Muğla      | 0.336 | Manavgat   | 0.677 | +0.341 [+0.253, +0.422] |
| <code>current_tvdi_mean</code>   | Evia       | 0.362 | Manavgat   | 0.677 | +0.315 [+0.221, +0.413] |
| <code>current_tvdi_mean</code>   | Montiferru | 0.356 | Manavgat   | 0.677 | +0.322 [+0.161, +0.453] |
| <code>downscaled_1st_mean</code> | Muğla      | 0.307 | Manavgat   | 0.683 | +0.376 [+0.295, +0.451] |
| <code>downscaled_1st_mean</code> | Evia       | 0.377 | Manavgat   | 0.683 | +0.307 [+0.209, +0.405] |
| <code>fused_1st_mean</code>      | Muğla      | 0.325 | Manavgat   | 0.666 | +0.341 [+0.261, +0.413] |
| <code>fused_1st_mean</code>      | Evia       | 0.376 | Manavgat   | 0.666 | +0.291 [+0.198, +0.386] |

There are fourteen reversals between pairs, across **seven** features, and thirteen of them involve Manavgat. Under the original label there were three, across two features: elevation and the LST anomaly. The feature-removal analysis of Section 3.11 keeps these two and does not re-select. Twenty-six further pairs reverse at the point estimate only, and they are not counted. The strict criterion reduces the number of findings: a difference interval on the pair (the test of

Section S1.12) would support seventeen further reversals. None of the fourteen is corrected for multiplicity; Table S30 gives the corrected counts.

## S2.1 The transfer matrix in precision-recall terms

ROC-AUC is used in the main text so that the results can be compared with the susceptibility literature. A susceptibility map is used to rank areas, so precision-recall measures its practical value. At target prevalences of 7.0 to 28.7 %, the two measures can differ strongly. Values are read from the step9b exports of the re-frozen outputs (paper/labelfix\_rerun/round5/tables/corrected/\*/step9b\_metrics.json).

**Table S11. Thermal transfer, PR-AUC against the no-skill baseline.** The baseline is the burned prevalence of the target. Lift is PR-AUC divided by this baseline; a lift of 1 is no better than a random ranking. Ordered by lift. Checked row by row by paper/code/appendix\_tables.py. PR-AUC intervals at 2-cell and 10-cell blocking for every direction are in paper/labelfix\_rerun/round7/r7a\_pr\_auc\_10cell.csv.

| Direction             | ROC-AUC | PR-AUC | No-skill | Lift        |
|-----------------------|---------|--------|----------|-------------|
| Evia → Manavgat       | 0.677   | 0.321  | 0.143    | <b>2.24</b> |
| Manavgat → Evia       | 0.654   | 0.407  | 0.287    | 1.42        |
| Bejís → Montiferru    | 0.594   | 0.289  | 0.212    | 1.36        |
| Evia → Montiferru     | 0.647   | 0.283  | 0.212    | 1.34        |
| Bejís → Muğla         | 0.618   | 0.093  | 0.070    | 1.33        |
| Muğla → Evia          | 0.653   | 0.379  | 0.287    | 1.32        |
| Montiferru → Bejís    | 0.548   | 0.093  | 0.072    | 1.28        |
| Montiferru → Muğla    | 0.619   | 0.089  | 0.070    | 1.27        |
| Evia → Muğla          | 0.577   | 0.086  | 0.070    | 1.23        |
| Muğla → Bejís         | 0.583   | 0.088  | 0.072    | 1.22        |
| Manavgat → Montiferru | 0.518   | 0.243  | 0.212    | 1.15        |
| Montiferru → Evia     | 0.586   | 0.317  | 0.287    | 1.11        |
| Muğla → Montiferru    | 0.531   | 0.214  | 0.212    | 1.01        |
| Manavgat → Muğla      | 0.438   | 0.060  | 0.070    | <b>0.85</b> |
| Evia → Bejís          | 0.448   | 0.059  | 0.072    | <b>0.82</b> |
| Montiferru → Manavgat | 0.404   | 0.113  | 0.143    | <b>0.79</b> |
| Bejís → Evia          | 0.383   | 0.226  | 0.287    | <b>0.79</b> |
| Manavgat → Bejís      | 0.396   | 0.054  | 0.072    | <b>0.75</b> |
| Muğla → Manavgat      | 0.345   | 0.101  | 0.143    | <b>0.70</b> |
| Bejís → Manavgat      | 0.314   | 0.098  | 0.143    | <b>0.68</b> |

| Direction   | ROC-AUC      | PR-AUC       | No-skill     | Lift        |
|-------------|--------------|--------------|--------------|-------------|
| <b>Mean</b> | <b>0.527</b> | <b>0.181</b> | <b>0.157</b> | <b>1.13</b> |

Seven directions are below their own no-skill baseline, and only one is above twice its baseline. The seven are the same seven that are below chance on ROC-AUC, as expected for a reversed ranking. At 1 km blocking, all seven have PR-AUC intervals fully below the baseline. At 5 km blocking two do: Bejís to Manavgat [0.067, 0.132] and Muğla to Manavgat [0.071, 0.135], against 0.143. Section 4.4 shows that the number of directions below chance depends largely on the study areas.

## S2.2 The same-geography event pair, in full

Section S1.13 reports this comparison. Section 4.4 shows that its elevation reversal is caused by the study area, and Section S1.13 explains why its thermal channels are not evaluated. The values per feature are kept here because this comparison led to the frame test.

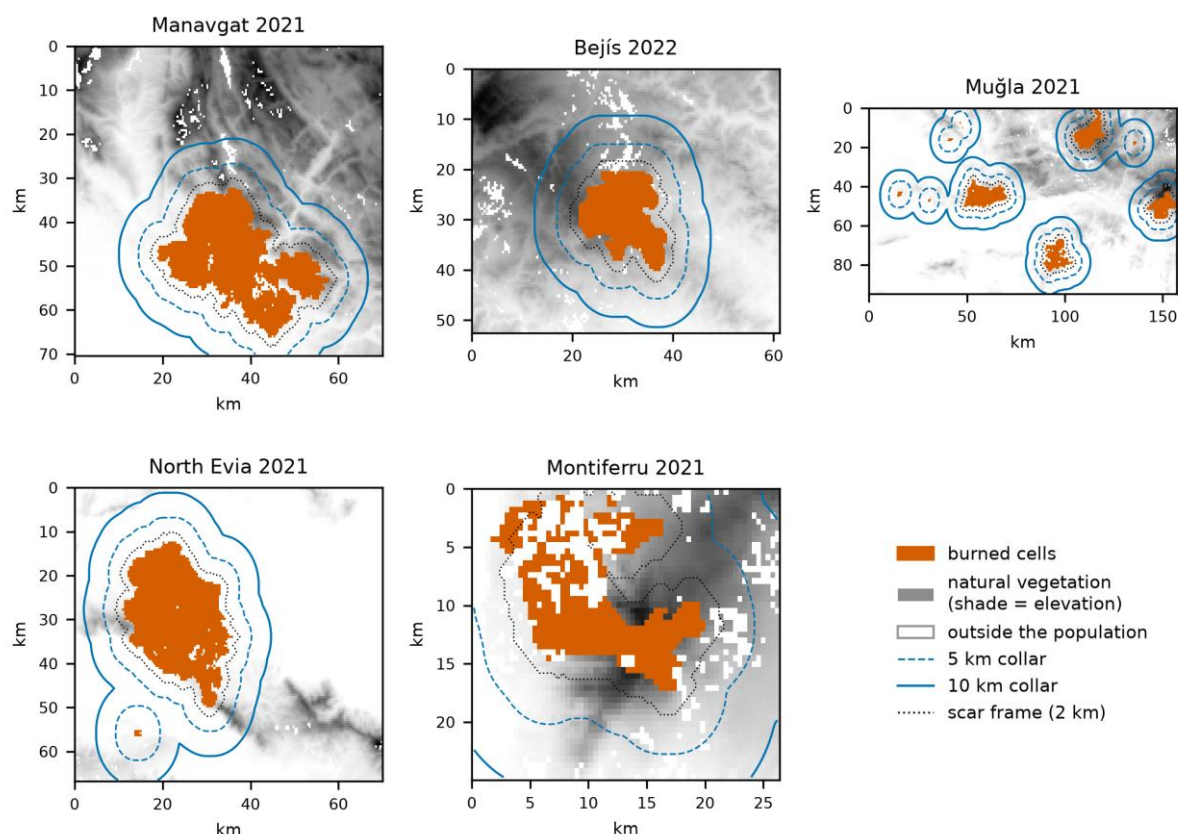
**Table S12. Signed univariate feature-burned AUC, Muğla 2021 versus 2022.** Raw AUC against burned, not folded to  $\max(\text{AUC}, 1 - \text{AUC})$ ; 10-cell ( $\approx 5$  km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.14). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Blocks of 5 km with burned cells: 70 in 2021 and 11 in 2022.** The note to Table 1 sets sixteen as the minimum for this blocking, so the 2022 intervals are indicative, like the 20-cell row of Table 1. The 2022 fire is also one compact scar, so its eleven blocks are next to each other. Read from paper/step9g\_raw/.../mugla\_2021\_\_mugla\_2022\_event\_relative/step9g\_direction\_reversal\_table.csv.

| Feature               | 2021 AUC [95 % CI]          | 2022 AUC [95 % CI]          | Reversal                   |
|-----------------------|-----------------------------|-----------------------------|----------------------------|
| <b>elevation_mean</b> | <b>0.611 [0.532, 0.690]</b> | <b>0.296 [0.230, 0.355]</b> | <b>bootstrap-supported</b> |
| current_lst_mean      | 0.325 [0.271, 0.382]        | 0.515 [0.434, 0.580]        | point only                 |
| current_tvdi_mean     | 0.336 [0.275, 0.398]        | 0.594 [0.475, 0.674]        | point only                 |
| downscaled_lst_mean   | 0.307 [0.253, 0.366]        | 0.508 [0.435, 0.571]        | point only                 |
| fused_lst_mean        | 0.325 [0.272, 0.383]        | 0.519 [0.436, 0.583]        | point only                 |
| ndvi_mean             | 0.662 [0.616, 0.704]        | 0.707 [0.624, 0.777]        | none                       |

| Feature              | 2021 AUC [95 % CI]   | 2022 AUC [95 % CI]   | Reversal |
|----------------------|----------------------|----------------------|----------|
| slope_mean           | 0.637 [0.582, 0.686] | 0.558 [0.468, 0.634] | none     |
| lst_anomaly_mean     | 0.485 [0.395, 0.566] | 0.380 [0.249, 0.502] | none     |
| tvdi_difference_mean | 0.490 [0.396, 0.575] | 0.397 [0.265, 0.514] | none     |

## S2.3 The evaluation frames, and the signed associations they produce

Section 4.4 gives these results. The values per region are given here.



**Fig. S1. The five study areas at cell level.** Grey shading is the elevation of the natural-vegetation cells, white cells lie outside this population, and burned cells are orange. The dashed and solid blue lines are the 5 km and 10 km distance collars, and the dotted black lines the 2 km scar frames of Table 2 (Section 3.12). In Muğla one burned area reaches the eastern edge of the study area (15 burned cells on the edge), so its collars are cut there. Axes give approximate distances in km on the ~500 m grid. Drawn by `paper/figures/figS1_study_areas.py` from the released modelling datasets.

696 **Table S13. Geometry of the five study areas.** Natural-vegetation population. Distance is the  
697 Euclidean distance to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Corrected  
698 Manavgat label. Computed by `paper/code/appendix_tables.py` from the step8a tables (read  
699 through `paper/code/_canonical.py`, which checks the sha256 of each file), as in  
700 `paper/code/verify_aoi_frame.py`.

| Region     | cells  | burned | median distance to burned | share beyond 10 km |
|------------|--------|--------|---------------------------|--------------------|
| Manavgat   | 20,511 | 2,935  | 13.1 km                   | <b>58.5 %</b>      |
| Bejís      | 15,190 | 1,100  | 13.5 km                   | <b>63.1 %</b>      |
| Muğla      | 41,730 | 2,911  | 11.3 km                   | 55.3 %             |
| Evia       | 9,298  | 2,664  | 8.0 km                    | 43.7 %             |
| Montiferru | 2,544  | 539    | 2.7 km                    | <b>2.1 %</b>       |

701 In Manavgat, the median elevation of the modelled cells is 330 m within 5 km of the fire  
702 (burned cells included), 995 m at 10 to 20 km and 1,273 m at 20 to 50 km; the burned cells  
703 have a median of 287 m (printed by `paper/code/verify_aoi_frame.py`, stage 1b).

704 **Table S14. Signed univariate AUC, original study area against a 10 km collar.** Point  
705 estimates. The intervals used for the reversal test are given in the text and in  
706 `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and not  
707 folded to  $\max(\text{AUC}, 1 - \text{AUC})$ , so a value below 0.5 shows a direction, not weakness. The  
708 collar removes no burned cells in any region. Full frame from the source of Table S9; collar  
709 from `paper/labelfix_rerun/code/aoi_frame_auc_frozen_mugla.csv` (the collar analysis  
710 with Muğla read from its canonical file). Checked row by row by  
711 `paper/code/appendix_tables.py`. On every row, Manavgat is on the other side of 0.5 from  
712 the other regions, also on the collar.

| Signed univariate AUC            | Manavgat     | Bejís | Muğla | Evia  | Montiferru | straddles 0.5 |
|----------------------------------|--------------|-------|-------|-------|------------|---------------|
| elevation, full frame (Table S9) | <b>0.232</b> | 0.643 | 0.611 | 0.541 | 0.584      | <b>yes</b>    |
| elevation, 10 km collar          | <b>0.376</b> | 0.614 | 0.606 | 0.648 | 0.581      | <b>yes</b>    |
| current LST, full frame          | <b>0.665</b> | 0.477 | 0.325 | 0.377 | 0.370      | <b>yes</b>    |
| current LST, 10 km collar        | <b>0.522</b> | 0.405 | 0.332 | 0.286 | 0.376      | <b>yes</b>    |
| current TVDI, full frame         | <b>0.677</b> | 0.517 | 0.336 | 0.362 | 0.356      | <b>yes</b>    |
| current TVDI, 10 km collar       | <b>0.527</b> | 0.454 | 0.342 | 0.250 | 0.361      | <b>yes</b>    |

**Table S15. Region summary: populations and gate results.** Counts from the Step 8A dataset statistics of each region; gate fractions from the burned-landcover gate output of each region. TSG = the natural-vegetation population (tree, shrub and grass). The TSG columns use the modelled population, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, on which every model in this paper was fitted and scored. Counts are computed from the step8a tables, and gate fractions are read from the gate output of each region (`paper/labelfix_rerun/round5/tables/corrected/gates/`). Checked row by row by `paper/code/appendix_tables.py`.

| Region                     | Total cells | Valid cells | Burned | Prevalence (all valid) | TSG cells | Burned in TSG | TSG prevalence | Burned natural-veg fraction | Gate verdict |
|----------------------------|-------------|-------------|--------|------------------------|-----------|---------------|----------------|-----------------------------|--------------|
| Manavgat 2021              | 24,150      | 24,087      | 3,046  | 0.126                  | 20,511    | 2,935         | 0.143          | 0.955                       | pass         |
| Bejís 2022                 | 15,759      | 15,759      | 1,103  | 0.070                  | 15,190    | 1,100         | 0.072          | 0.991                       | pass         |
| Muğla 2021                 | 73,098      | 73,045      | 3,026  | 0.041                  | 41,730    | 2,911         | 0.070          | 0.958                       | pass         |
| North Evia 2021 (extended) | 22,925      | 22,906      | 2,788  | 0.122                  | 9,298     | 2,664         | 0.287          | 0.945                       | pass         |
| Montiferru 2021            | 3,234       | 3,173       | 697    | 0.220                  | 2,544     | 539           | 0.212          | 0.723                       | pass         |

**Table S16. Cross-region transfer matrix, thermal model, TSG population.** Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ( $\lambda = 10^{-5}$ ). Generated by `paper/code/table_b9.py` from the re-frozen outputs (`paper/labelfix_rerun/round5/matrix20_official.csv`).

| Direction      | Raw                  | Region-wise z-score  | CORAL                |
|----------------|----------------------|----------------------|----------------------|
| Manavgat→Bejís | 0.396 [0.373, 0.422] | 0.450 [0.423, 0.477] | 0.467 [0.441, 0.491] |
| Bejís→Manavgat | 0.314 [0.296, 0.332] | 0.302 [0.282, 0.323] | 0.406 [0.388, 0.423] |
| Manavgat→Muğla | 0.438 [0.418, 0.456] | 0.427 [0.408, 0.444] | 0.417 [0.398, 0.436] |
| Muğla→Manavgat | 0.345 [0.331, 0.359] | 0.485 [0.468, 0.502] | 0.476 [0.460, 0.493] |
| Manavgat→Evia  | 0.654 [0.633, 0.676] | 0.529 [0.504, 0.553] | 0.504 [0.481, 0.528] |
| Evia→Manavgat  | 0.677 [0.658, 0.697] | 0.404 [0.386, 0.421] | 0.417 [0.399, 0.435] |
| Bejís→Muğla    | 0.618 [0.601, 0.635] | 0.518 [0.501, 0.535] | 0.507 [0.489, 0.524] |
| Muğla→Bejís    | 0.583 [0.561, 0.607] | 0.535 [0.512, 0.557] | 0.560 [0.538, 0.581] |
| Bejís→Evia     | 0.383 [0.363, 0.402] | 0.532 [0.509, 0.551] | 0.499 [0.479, 0.518] |
| Evia→Bejís     | 0.448 [0.426, 0.470] | 0.549 [0.524, 0.575] | 0.549 [0.525, 0.573] |
| Muğla→Evia     | 0.653 [0.636, 0.671] | 0.561 [0.543, 0.580] | 0.563 [0.545, 0.582] |

| Direction           | Raw                  | Region-wise z-score  | CORAL                |
|---------------------|----------------------|----------------------|----------------------|
| Evia→Muğla          | 0.577 [0.560, 0.593] | 0.501 [0.485, 0.518] | 0.530 [0.515, 0.546] |
| Montiferru→Manavgat | 0.404 [0.385, 0.422] | 0.388 [0.367, 0.408] | 0.436 [0.416, 0.456] |
| Manavgat→Montiferru | 0.518 [0.474, 0.563] | 0.527 [0.484, 0.568] | 0.505 [0.461, 0.547] |
| Montiferru→Bejís    | 0.548 [0.521, 0.578] | 0.574 [0.552, 0.596] | 0.569 [0.548, 0.591] |
| Bejís→Montiferru    | 0.594 [0.560, 0.631] | 0.550 [0.500, 0.601] | 0.574 [0.530, 0.621] |
| Montiferru→Muğla    | 0.619 [0.604, 0.634] | 0.576 [0.562, 0.589] | 0.565 [0.550, 0.579] |
| Muğla→Montiferru    | 0.531 [0.495, 0.568] | 0.587 [0.549, 0.624] | 0.584 [0.547, 0.623] |
| Montiferru→Evia     | 0.586 [0.565, 0.606] | 0.630 [0.611, 0.649] | 0.624 [0.605, 0.641] |
| Evia→Montiferru     | 0.647 [0.608, 0.682] | 0.568 [0.528, 0.609] | 0.581 [0.539, 0.623] |

## S2.4 The transferability-diagnostics tables

**Table S17. All similarity measures against raw thermal transfer (20 ordered directions).**

Spearman  $\rho$  with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with  $n = 12$  exist only for the four-region subset, because these measures were not produced for Montiferru. The supported conditional rows use the 18 directions with at least one supported feature. Every row is from `paper/labelfix_rerun/round5/s6_diagnostics_20.csv`, checked by `paper/code/appendix_tables.py`. The twenty measures were defined on 8 August 2026 and were not re-selected after the label correction.

| Diagnostic                             | Family                | Exp. | n<br>dir | Spearman $\rho$<br>[95% CI] | CI excludes 0 |
|----------------------------------------|-----------------------|------|----------|-----------------------------|---------------|
| Agreement fraction, supported features | P(y x)<br>conditional | +    | 18       | +0.52 [−0.27, +0.87]        | no            |
| Cosine, supported features             | P(y x)<br>conditional | +    | 18       | +0.49 [−0.24, +0.87]        | no            |
| Cosine, all 9 features                 | P(y x)<br>conditional | +    | 20       | +0.44 [−0.29, +0.80]        | no            |
| Vector Spearman, all 9                 | P(y x)<br>conditional | +    | 20       | +0.43 [−0.21, +0.80]        | no            |
| Agreement count, all 9                 | P(y x)<br>conditional | +    | 20       | +0.47 [−0.29, +0.80]        | no            |
| Schoener's D, 1-D mean                 | P(x y=1)<br>niche     | +    | 20       | +0.10 [−0.53, +0.63]        | no            |
| Warren's I, 1-D mean                   | P(x y=1)<br>niche     | +    | 20       | +0.19 [−0.43, +0.77]        | no            |
| Schoener's D, PCA-2D                   | P(x y=1)<br>niche     | +    | 20       | +0.03 [−0.59, +0.63]        | no            |
| Warren's I, PCA-2D                     | P(x y=1)<br>niche     | +    | 20       | −0.03 [−0.66, +0.56]        | no            |
| Mahalanobis, burned centroids          | P(x y=1)<br>niche     | −    | 20       | −0.22 [−0.82, +0.42]        | no            |

| Diagnostic                                        | Family                | Exp. | n<br>dir | Spearman $\rho$<br>[95% CI] | CI excludes 0                           |
|---------------------------------------------------|-----------------------|------|----------|-----------------------------|-----------------------------------------|
| Domain-classifier AUC                             | P(ix)<br>marginal     | –    | 20       | –0.33 [–0.81,<br>+0.36]     | no                                      |
| Predictor-space mean<br>dissimilarity             | P(ix)<br>marginal     | –    | 12       | –0.13 [–0.67,<br>+0.37]     | no                                      |
| Predictor-space p95<br>dissimilarity              | P(ix)<br>marginal     | –    | 12       | –0.15 [–0.68,<br>+0.24]     | no                                      |
| Fraction inside weighted AoA                      | P(ix)<br>marginal     | +    | 12       | +0.21 [–0.48,<br>+0.62]     | no                                      |
| Fraction inside unweighted<br>support             | P(ix)<br>marginal     | +    | 12       | +0.08 [–0.92,<br>+0.76]     | no                                      |
| Climatic distance                                 | P(ix)<br>marginal     | –    | 12       | +0.03 [–0.84,<br>+0.84]     | no                                      |
| Geographic distance                               | geographic            | –    | 12       | –0.17 [–0.87,<br>+0.84]     | no                                      |
| Regime distance, log effective-<br>N              | P(y) structure        | –    | 20       | +0.11 [–0.67,<br>+0.80]     | no                                      |
| Regime distance, largest share                    | P(y) structure        | –    | 20       | +0.11 [–0.66,<br>+0.77]     | no                                      |
| Vector Spearman, supported ( $\geq 3$<br>feats) ‡ | P(y x)<br>conditional | +    | 6        | +0.96 [+0.85,<br>+0.96]     | degenerate interval; not<br>interpreted |

*Table note.* A null row means that the measure was **not shown to order transfer** in this design. It does not mean that the measure cannot order transfer. With ten independent region pairs the power is low, and the intervals are wide enough to include moderate true correlations in either direction. ‡ The last row belongs to the set of measures fixed in advance, so it stays in the table. It is defined on six directions only, and its interval is degenerate (the upper bound equals the point estimate), so it is not interpreted and cannot be compared with the other nineteen.

**Equal-sample check.** The families use different samples: twelve directions for the marginal, applicability, climatic and geographic rows, eighteen for the supported conditional rows, and twenty for the others. Every row was recomputed on the common subset of twelve directions, and again on the sixteen directions where the supported conditional rows are defined. Every interpretable row still includes zero. On the common twelve, the supported conditional rows give +0.41 [–0.42, +0.88] and +0.46 [–0.56, +0.84]. The published values are reproduced to

746  $4.4 \times 10^{-5}$ . Source:  
747 `paper/labelfix_rerun/round5/out_official/diagnostics_common_subset.json`.

748 **Table S18. The most and least environmentally similar pairs, on both frames.** Schoener's  
749  $D$  is computed over burned cells only, so it does not change with the collar. Transfer values  
750 are the two ordered directions of each pair. Ranks are out of the twenty directions on the  
751 collar. Corrected Manavgat label. Schoener's  $D$ , per-feature  $D$  and transfer on the original  
752 study areas are read from the source of Fig. 8,  
753 `paper/labelfix_rerun/round5/out_official/figure_contrast_pairs.json`. Collar  
754 transfer and ranks are from the source of Table S8,  
755 `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv`, and the AoA shares  
756 from `paper/labelfix_rerun/round5/collar/aoa_directed_pair_summary.csv`. Every cell  
757 is checked against these files at build time (`paper/figures/fig8_contrast_pairs.py`).

|                                           | Manavgat and Muğla     | Bejís and Montiferru  |
|-------------------------------------------|------------------------|-----------------------|
| Schoener's $D$ , mean 1-D                 | <b>0.799</b> (highest) | <b>0.479</b> (lowest) |
| per-feature $D$                           | 0.74 to 0.87           | 0.23 to 0.77          |
| transfer, frames as drawn                 | 0.438, 0.345           | 0.594, 0.548          |
| transfer, 10 km collar                    | 0.493, 0.433           | 0.669, 0.624          |
| rank of 20 on the collar, from the bottom | 4th, 2nd               | 16th, 12th            |
| target cells inside the AoA               | 0.876, 0.531           | n/a                   |

## 758 S3 Protocol detail and limitations

759 These protocols are given here in full. They are needed only to check the related results, not to  
760 follow the main argument. Sections 3.2 to 3.4, 3.7, 3.9, 3.10, 3.13 and 3.14 summarise them.

### 761 S3.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate

762 Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1), retrieved  
763 through Google Earth Engine (Gorelick et al. 2017). Its omission and commission errors  
764 (Boschetti et al. 2019) limit every model here. The product is exported onto the 30 m

765 EPSG:4326 reference grid used by the rest of the pipeline. Each native ~500 m observation is  
766 therefore copied to the 30 m pixels below it.

767 The analysis grid is therefore **reconstructed, not native**. The 30 m grid is divided into non-  
768 overlapping blocks of  $\text{round}(500 / 30) = 17 \times 17$  pixels, which gives a nominal cell edge  
769 of 510 m. Each cell is identified by its block indices `row_500m` and `col_500m`. This  
770 approximates the MODIS sinusoidal cell but does not reproduce it, because it is anchored to  
771 the Landsat/EPSG:4326 reference grid. Blocks at the edges are cut. The cell is square in  
772 degrees but not on the ground: about 510 m from north to south and 390 to 407 m from east to  
773 west. Its area is therefore 0.199 to 0.208 km<sup>2</sup>, against 0.215 km<sup>2</sup> for the native MODIS cell  
774 (463 m edge). Every block size in this paper refers to the north-south dimension. The collar  
775 step of 0.45 km per cell (Section 3.12) lies between the two cell dimensions, so collar radii are  
776 nominal to within about 15 %.

777 The burn date of a cell is the **mode** of the burned sub-pixel day-of-year values in the block,  
778 and it is tested against the label window of the region. The exported raster has no burned  
779 pixels outside the window, so in this dataset one burned sub-pixel inside the window makes  
780 the cell burned. This can extend each scar by up to one cell at its edge, and it can mark a water  
781 cell as burned where the product does. The primary population keeps only cells dominated by  
782 natural vegetation, so water is excluded. The fraction of burned sub-pixels that agree with the  
783 modal date is recorded as `burn_date_pixel_agreement_fraction`, but no threshold is  
784 applied. The label never affects whether a cell is modelled: unburned, all-no-data and out-of-  
785 window cells all stay in the negative class.

786 Earlier burning is handled differently in each region; this is given in Section S3.1.1. The only  
787 historical exclusion applied by design removes the 2021 Muğla scar from the 2022  
788 experiment of Section 3.14.

Before any modelling, each region passes a gate that asks one question: what fraction of the burned cells is dominated by natural vegetation? A region is admitted as a wildfire case when this fraction is at least 0.50 and at least 30 burned cells are present. It is rejected as a cropland control when the cropland fraction is at least 0.50. The gate uses ESA WorldCover classes aggregated to the same cells. It separates burning of natural fuel from stubble burning after harvest, which MCD64A1 does not distinguish. The results are given in Section 4.1.

### S3.1.1 Pre-label and historical burning, by region

Two kinds of earlier burning can affect a label. **Pre-label burning** is burning inside the predictor window of the region, which the pipeline can remove. **Historical burning** is burning in the five previous years, which no region screens for. Table S34 gives both, recomputed from the MCD64A1 export of each region (paper/code/ems\_labels\_1\_prelabel.py, ems\_labels\_2\_history.py; outputs paper/labelfix\_rerun/labels/r1\_prelabel.json, r2\_history.json).

**Table S34. Earlier burning by region, natural-vegetation population.**

| Region          | Pre-label cells removed            | Pre-label burned cells retained             | Historical-burn cells (five years) | ... of which burned again in the event |
|-----------------|------------------------------------|---------------------------------------------|------------------------------------|----------------------------------------|
| Manavgat 2021   | none needed (no pre-label burning) | 0                                           | 0                                  | 0                                      |
| Bejís 2022      | exclusion not applied              | 48 (49 in all valid cells; 10 burned again) | 163                                | 0                                      |
| Muğla 2021      | 49                                 | 0                                           | 194                                | 25                                     |
| North Evia 2021 | 16                                 | 0                                           | 366                                | 88                                     |
| Montiferru 2021 | 61                                 | 0                                           | 58                                 | 32                                     |

Excluding the historically burned cells lowers the mean paired transfer contribution from +0.007 to +0.003 (at 10-cell blocking: 5 positive, 4 negative and 11 uncertain verdicts, against 6, 5 and 9). The within-region gain stays positive with interval support in every region:

+0.062 (no change), +0.055, +0.088, +0.162 and +0.111 for Manavgat, Bejís, Muğla, Evia and Montiferru.

### **S3.1.2 The Manavgat label correction**

**The Manavgat 2021 label is corrected.** The label first used for Manavgat was exported on 8 July 2026. This was before the month-alignment fix of the MCD64A1 query in the pipeline (commit 183be42, 11 July), and the export was not renewed. It therefore missed the first four days of the fire, 28 to 31 July. The error was an old export, not a code error. The corrected label only adds burned cells. No cell that burned under the original label becomes unburned, and no predictor or validity flag changes. In the primary population, the number of burned cells rises from 784 to 2,935. The Manavgat outputs were re-frozen with the pipeline at commit 6381f4c. Two small changes were made: a one-line patch to the window-closure diagnostic, and a runtime setting of the Earth Engine project (thermaltwin). Python 3.12.10 and scikit-learn 1.9.0 were used (versions in `ENVIRONMENT.md`). The correction, the re-freeze and the control run were carried out by the corresponding author with AI-assisted code (Section 3.13). They were not repeated by the author of the pipeline. This pipeline version writes one extra column, a flag for burning in earlier years, and it excludes no Manavgat cell. In a control run, the original label was re-frozen with the same code and environment. It reproduced the published outputs to within  $10^{-5}$ . The exceptions are listed in the manifest: file paths, the column list, and one older pair that differs at the level of random-forest thread nondeterminism. Differences between the two runs are therefore caused by the label alone. The reproduction check of Section 3.13 covers the re-frozen outputs. The manifest, hashes and scripts are released in `paper/data/manavgat_2021/refreeze/`.

### S3.1.3 Effect on Table 1

The Manavgat rows use the corrected label. It adds burned cells (784 to 2,935 in the primary population) and changes no predictor (Section 3.2). The rows become stronger, not weaker. Absolute AUCs rise by 0.04 to 0.11 across the three block sizes, and the gain holds at every scale. The 1 km interval becomes narrower, from  $[+0.055, +0.079]$  to  $[+0.060, +0.073]$ , because the region now has 814 blocks of 2 cells with burned cells instead of 235.

### S3.2 Transferability diagnostics versus transfer

Twenty similarity measures from five families are each rank-correlated with the same quantity, the raw thermal transfer AUC over the twenty ordered directions, with one common pair-based bootstrap. The families are marginal predictor-distribution measures  $P(x)$ , burned-niche overlap  $P(x|y=1)$ , burned-pattern structure  $P(y)$ , and conditional feature-response direction  $P(y|x)$ .

The **marginal** family includes area-of-applicability dissimilarity in scaled, importance-weighted predictor space (Meyer and Pebesma 2021, 2022; Ludwig et al. 2023), climatic and geographic distance, and a domain classifier trained to separate source cells from target cells. The **niche-overlap** family includes Schoener's D (Schoener 1968) and Warren's I (Warren et al. 2008) over the burned cells of each region. The **regime** family includes counts of burned patches and effective patch counts. The **conditional** family is the sign-agreement index: the fraction of predictors whose signed association points the same way in both regions, computed over all predictors and over the predictors with interval support.

Two properties of this design limit what it can show. First, the set of measures was recorded on 8 August 2026, together with its results under the original label (commit abd6a4b on the history-archive branch of the public repository, which keeps the development history without internal notes). It was not re-selected after the label correction. This is a record, not a

pre-registration. Second, with five regions the effective sample is ten unordered pairs, and nineteen measures are evaluated at this power without family-wise error control. Where a measure passes an interval test but not a Bonferroni threshold, both results are reported.

### S3.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the only region in which a second fire is analysed on the same study area and grid. This allows the direction of feature-label associations to be compared with the place held fixed. The 2022 windows are set relative to the 2022 fire: a 58-day predictor window that closes the day before ignition, and a 49-day label window that opens on that day. These lengths are the same as in 2021.

**This is not a clean design for transfer over time, and it is not presented as one.** The 2022 fire started about six weeks earlier in the season than the 2021 fire, so year and seasonal phase are confounded, and no difference can be attributed to the year alone. The comparison is between two fires in one place. The pair is kept out of the twenty-direction matrix, and this is enforced in the released code.

The 2022 population is the 2021 population without the 2021 scar; Section S1.13 describes what follows from this.

**Table S19. Study regions, study areas and time windows.** Bounding boxes are in EPSG:4326, as recorded. Window lengths in brackets are day counts, both ends included. The baseline years are the four years before each predictor window, with the same calendar dates.

| Region                  | Bounding box (lon min, lat min, lon max, lat max) | Predictor window                | Label window                    | Baseline years         |
|-------------------------|---------------------------------------------------|---------------------------------|---------------------------------|------------------------|
| Manavgat 2021 (Türkiye) | 31.05, 36.72, 31.85, 37.35                        | 2021-06-01 to 2021-07-27 (57 d) | 2021-07-28 to 2021-08-31 (35 d) | 2017, 2018, 2019, 2020 |
| Bejís 2022 (Spain)      | -1.05, 39.68, -0.35, 40.15                        | 2022-06-15 to 2022-08-14 (61 d) | 2022-08-15 to 2022-09-30 (47 d) | 2018, 2019, 2020, 2021 |
| Muğla 2021 (Türkiye)    | 27.1, 36.6, 28.9, 37.45                           | 2021-06-01 to 2021-07-28 (58 d) | 2021-07-29 to 2021-09-15 (49 d) | 2017, 2018, 2019, 2020 |

| Region                      | Bounding box (lon min, lat min,<br>lon max, lat max) | Predictor window                   | Label window                       | Baseline<br>years         |
|-----------------------------|------------------------------------------------------|------------------------------------|------------------------------------|---------------------------|
| North Evia<br>2021 (Greece) | 23.05, 38.55, 23.85, 39.15                           | 2021-06-05 to<br>2021-08-02 (59 d) | 2021-08-03 to<br>2021-09-30 (59 d) | 2017, 2018,<br>2019, 2020 |
| Montiferru<br>2021 (Italy)  | 8.45, 40.05, 8.75, 40.27                             | 2021-05-25 to<br>2021-07-23 (60 d) | 2021-07-24 to<br>2021-08-31 (39 d) | 2017, 2018,<br>2019, 2020 |

### S3.4 Predictor provenance, compositing and the downscaler

Ten predictors are used, four baseline and six thermal. All are summarised per cell over the predictor window, through the chain shown in Fig. 2. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1\_L2). They are quality-screened per pixel with the QA\_PIXEL band; the water bit is kept on purpose. The coarse thermal input is MODIS/061/MOD11A1.

- **NDVI** is the median of Landsat surface reflectance over the predictor window. It is the only baseline predictor that changes over time.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30. Its heights are referenced to the EGM2008 geoid (EPSG:3855) in metres, according to the product handbook. The pipeline applies no datum conversion.
- **Land cover** is ESA WorldCover v200 (Zanaga et al. 2022), used as the dominant class code.
- **Current LST** is the median of Landsat surface temperature over the predictor window, in °C.
- **LST anomaly** is a z-score of the current LST median against the four baseline years. It is set to no-data where the baseline standard deviation is below 1.0 °C or there are too few observations.
- **TVDI** and **TVDI difference**. The Temperature-Vegetation Dryness Index (Sandholt et al. 2002) is computed in the LST-NDVI space. The wet and dry edges are the 2nd and 98th LST percentiles within each of twenty NDVI bins, and the index is limited to [0,

1]. The edges are percentiles of the LST values in a given scene, so they are fitted for each study area and window. A TVDI of 0.5 therefore means a different moisture state in each region. **TVDI difference** is the raw anomaly of this index against the same four baseline years: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`. It is in index units, not standard deviations, so it is reported together with the z-scored channel.

- **Downscaled LST and fused LST.** A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals the observed Landsat LST where that is valid, and the downscaled surface only where it is not. Gap-filling therefore never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler uses coordinates as inputs. This is the only way in which coordinate information enters a feature set from which Section 3.13 excludes coordinates. Section S1.7 gives the gain without these two channels.

**The six thermal channels carry about two signals.** In the primary population, current, fused and downscaled LST are correlated at 0.97 to 1.00 in every region. Current LST and TVDI are correlated at 0.88 to 0.98, and the two anomaly channels at 0.71 to 0.94. Two principal components of the six channels explain 92 % to 98 % of their variance, and their participation ratio is 1.6 to 1.8 (`paper/labelfix_rerun/labels/r4_dimensionality.json`).

**Table S20. Input data.** All retrieved through Google Earth Engine.

| Input                               | Product and version                                                                              | Native resolution | Use                              | Reference                                                                           |
|-------------------------------------|--------------------------------------------------------------------------------------------------|-------------------|----------------------------------|-------------------------------------------------------------------------------------|
| Burned area                         | MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1)                                                 | 500 m, monthly    | label                            | (Giglio et al. 2018, 2021)                                                          |
| Surface reflectance and temperature | Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), QA_PIXEL screening, water bit preserved | 30 m              | NDVI, current LST, anomaly, TVDI | (U.S. Geological Survey Earth Resources Observation and Science (EROS) Center 2020) |

| Input      | Product and version                     | Native resolution | Use                                                  | Reference                    |
|------------|-----------------------------------------|-------------------|------------------------------------------------------|------------------------------|
| Coarse LST | MODIS MOD11A1 v061 (MODIS/061/MOD11A1)  | 1 km, daily       | downscaled and fused LST (gap-fill 0.11 % to 9.70 %) | (Wan et al. 2021)            |
| Terrain    | Copernicus DEM GLO-30 (EGM2008 heights) | 30 m              | elevation, slope                                     | (European Space Agency 2022) |
| Land cover | ESA WorldCover v200 (2021)              | 10 m              | dominant class, gate, population                     | (Zanaga et al. 2022)         |

912 Compositing uses the median over the predictor window. The anomaly channels refer to four  
913 baseline years (Table S19). WorldCover 2021 is later than four of the five fires (Section 3.4).

914 **Table S36. Landsat 8 observations per window.** Scenes = Landsat 8 Collection 2 Level-2  
915 scenes whose footprint intersects the study area. Clear observations = number of clear  
916 observations per pixel after the pipeline's QA mask (`src/step3_landsat_1st.py`), median  
917 and 10th to 90th percentile over the study area at 300 m. The pipeline also filters the predictor  
918 window by calendar month, so its counts can be slightly lower. Landsat 9 is not used. Source  
919 `paper/labelfix_rerun/round8/r8k_landsat_observations.csv`.

| Region          | Predictor window scenes | Clear observations, predictor window | Clear observations, baseline years (medians) |
|-----------------|-------------------------|--------------------------------------|----------------------------------------------|
| Manavgat 2021   | 14                      | 6 [3 to 10]                          | 7, 7, 6, 6                                   |
| Bejís 2022      | 16                      | 3 [2 to 6]                           | 4, 4, 3, 2                                   |
| Muğla 2021      | 18                      | 6 [3 to 10]                          | 6, 6, 6, 6                                   |
| North Evia 2021 | 8                       | 4 [3 to 4]                           | 4, 3, 2, 1                                   |
| Montiferru 2021 | 4                       | 3 [2 to 4]                           | 4, 3, 2, 3                                   |

920 The anomaly channels therefore rest on few clear observations per year, one to seven  
921 depending on region and year, which adds noise to the four-year baseline.

## 922 S3.5 Limitations, in full

923 Section 5.6 lists the limitations that affect the conclusions. All fourteen are given here.

(i) **No weather predictors** are used, so the behaviour of a combined thermal and weather predictor set is not known.

(ii) **The same-place comparison covers one region only**, and there, year and seasonal phase are confounded. This cannot be resolved in this study area. The two fires are 42 days apart in median burn day-of-year, neither year has a second fire at the phase of the other, and a calendar-matched comparison would have nine burned cells, against a gate minimum of thirty. The 2022 population also has only eleven blocks with burned cells, against the minimum of sixteen (Section S1.13). Its 331 burned cells do not resolve the thermal reversals at interval level, and the pair keeps the place fixed but not the population. The two Muğla populations were computed with the unmodified pipeline code and the same fixed environment, not read from a frozen export.

(iii) **All labels come from one burned-area product**, MCD64A1 (Giglio et al. 2018). Its omission and commission errors (Boschetti et al. 2019) limit every model here. Its accuracy in the Mediterranean was assessed against reference perimeters (Katagis and Gitas 2022). A second 500 m product, VIIRS VNP64A1 (Giglio 2024), was used as a label check (Table S35), and no conclusion changed. Independent fire perimeters were not used.

(iv) **Evia has the most unusual class balance**, even on the extended study area. Its TSG prevalence is 0.287, the highest in the cohort, against 0.070 to 0.212 in the other four regions.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with the weather of each fire. Labels from several years are needed to separate them.

(vi) **Point estimates depend on the software version.** Under scikit-learn 1.5.2 instead of 1.9.0, single-direction AUCs of the frame-transfer matrix move by up to 0.023 for the thermal model and 0.047 for the baseline, and paired differences by up to 0.045. The means move by at most 0.003. One direction changes interval support on each of the original and 10 km frames (paper/labelfix\_rerun/round7/sklearn152/compare\_vs\_1\_9\_0.json). All reported numbers come from one verified version, and exact reproduction needs the archived environment.

(vii) **The low transfer into Manavgat is located but not explained.** Manavgat is the weakest target (0.435), has the largest matched shortfall (+0.319) and has the most unusual univariate profile in the cohort. Two processing causes were tested, and neither explains it. The quality screening of its coarse thermal input moves no signed association by more than 0.005 (Section S1.5), and its elevation reversal remains on the collar (Section 4.4). Holding elevation reverses its raw LST association (Section S1.11), while holding aspect does not. This rests on one region and one fire.

(viii) **Measures based on interval support are less stable than the point estimates.** Five to six of the pipeline's interval bounds lie within 0.01 of 0.5, under both labels. One such flag, the NDVI of Manavgat, moved the supported-feature cosine on the original study areas from  $p = 0.49$  to 0.70, when two equally valid bootstrap streams disagreed on it. The transfer verdicts are more stable. Across five bootstrap seeds, every verdict at 1 km blocking is stable, and the paired split of twelve positive, seven negative and one uncertain holds on every seed. At 5 km blocking, one level verdict and two paired-delta verdicts change with the seed. Exact counts of supported directions should be read at this precision.

(ix) **The five study areas are not comparable, and this cohort cannot fully correct this**

(Section 4.4). The collar results are reported together with the original results, not instead of them, because the collar radius is also a choice, and 5 km and 10 km do not agree exactly (0.591 against 0.589). The study areas were fixed earlier in the pipeline, in repo/, so they can be restricted but not extended. Montiferru, whose rectangle is already close to the fire, cannot be given distant cells. Their independence from the outcomes can also not be fully documented. Only the Montiferru rectangle follows a rule, and the Manavgat rectangle is dated before its first gate result. For Bejís and Muğla, the version history cannot show that the rectangle was fixed before the first gate result (Section 3.1). Future studies should fix the study area by a stated accessible-area rule (Barve et al. 2011) before any predictor is computed.

(x) **Other classifiers are compared by point estimate only.** The main results use a

random forest with unlimited depth, which fits local structure well and extrapolates poorly. Section S1.8 shows that the transfer result does not depend on this choice. Three other estimators, including a penalised linear model, range from 0.481 to 0.527 and have eleven to thirteen of twenty directions above chance. These are point estimates without intervals, so the estimators are not ranked. Other estimator classes were not tried.

(xi) **Burn timing and elevation are confounded in Manavgat.** The cells that burned in the first four days lie at a median of 219 m, against 512 m for the later burns and 1,004 m for unburned cells. The effects of timing and elevation cannot be separated in these data.

(xii) **The quality-screening comparison differs in code version as well as in screening.** The current pipeline step7 refuses the unscreened MODIS input, because

the raster has no nodata tag and 8.1 % exact zeros. The unscreened version therefore uses the step7 of export time, and the screened version the current one. The two agree on elevation to four decimals and on every other signed AUC to within 0.005; the largest change is  $-0.0044$  for downscaled LST. The within-region gain moves from  $+0.067$  to  $+0.068$ . The confound could hide an effect only if two effects cancelled (Section S1.5).

(xiii) **The similarity tests rest on ten independent region pairs.** Five regions give twenty ordered directions, but the two directions of a pair share both regions and are not independent. All results of Section S1.21, positive and negative, have this low power. With ten pairs, a moderate true ordering would usually be missed.

(xiv) **The frame cost rests on few scars.** The full comparison of Section 4.3 is defined on seven scars in three regions. Its scar-level intervals are too narrow, because row A is repeated for each scar of a region. The region-level estimates ( $+0.137$  over three regions,  $+0.160$  over five) should be quoted.

### **S3.6 Leakage control and reproducibility, in full**

A fixed set of forbidden columns is checked at every model fit. Coordinates (lon, lat, row, col and their normalised forms), every burn-date and label-provenance column, and the agreement fraction are excluded from all feature sets. The check is an assertion in the code. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking keeps neighbouring cells in the same fold.

**Reproducibility.** All random processes use seed 42, and every bootstrap uses 1000 replicates. The reproduction check is described in Section S3.6.2.

### 1016 S3.6.1 Resampling units

1017 Uncertainty is estimated with a spatial-block bootstrap. Let  $\beta_1, \dots, \beta_M$  be the blocks of Eq. (1)  
 1018 of the main text that contain cells of  $F$ . Replicate  $b$  draws  $M$  indices  $u_{bm}$  uniformly with  
 1019 replacement:

$$1020 \quad F^{*b} = \mathbb{U}_{m=1}^M \beta_{u_{bm}}, \quad \text{CI}_{95} = [Q_{0.025}\{\theta(F^{*b})\}_b, Q_{0.975}\{\theta(F^{*b})\}_b]. \quad (S1)$$

1021 Here  $\theta$  is the statistic and  $Q$  gives the 2.5 and 97.5 percentiles over 1000 replicates, seed 42.  
 1022 A replicate with only one class is dropped. Differences are formed within each replicate, so  
 1023 they are paired. The block size of each interval is given with the result. Because blocks are  
 1024 resampled, the reliability of an interval depends on the number of blocks with at least one  
 1025 burned cell. These counts are reported with the intervals. Where a verdict rests on too few  
 1026 such blocks, it is reported as indicative.

1027 The twenty transfer directions are not independent, because each region appears in eight of  
 1028 them. The **pair-cluster bootstrap** therefore resamples the ten unordered region pairs. Each  
 1029 pair carries its set  $\pi_p$  of ordered directions, and the carried values  $\delta_{st}$  are averaged:

$$1030 \quad \bar{\delta}^{*b} = \frac{\sum_{m=1}^{10} \sum_{(s,t) \in \pi_{u_{bm}}} \delta_{st}}{\sum_{m=1}^{10} |\pi_{u_{bm}}|}. \quad (S2)$$

1031 The interval has the same percentile form, with 1000 replicates for the direction-level  
 1032 intervals of Sections 4.4 and 4.5, and 20,000 for the unit comparison of Table S31. Clustering  
 1033 by target region replaces  $\pi_p$  by the four directions that share a target. A quantity with one  
 1034 value per held-out scar or target region gets a Student t interval over these  $n$  units:

$$1035 \quad \bar{x} \pm t_{0.975, n-1} s_x / \sqrt{n}. \quad (S3)$$

**The effective sample is therefore ten pairs or five regions for direction-level intervals, and at most seven scars from three regions for scar-level intervals.**

### **S3.6.2 Reproduction of the re-frozen outputs**

The transfer analysis runs in an environment separate from the pipeline. The reproduction check of the repository therefore refitted every within-region model and all twenty CORAL transfers and compared them with the frozen pipeline output. For Manavgat this output is the one re-frozen on the corrected label (Section 3.2), produced with the pipeline at commit 6381f4c. Only two changes were made. A one-line patch lets the window-closure module accept a population column that the corrected label adds; its diff is released with the re-freeze. A runtime setting points the Earth Engine calls at the project `thermaltwin`, which changes only the infrastructure. The within-region results agree exactly, and the transfer directions to within  $1.3 \times 10^{-8}$ , below the repository tolerance of  $10^{-6}$ . This tolerance applies to the CORAL and within-region check. The frame-transfer script of Section 4.4 fits its forests in parallel, so its values vary between runs by up to  $2 \times 10^{-6}$  ( $1.4 \times 10^{-6}$  in a rerun on 29 September 2026, `paper/labelfix_rerun/round7/`). This is within the tolerance of  $10^{-5}$  applied to it, and the values are stable at the printed precision. The effect of an unpinned library version is given in Section S3.5(vi).

### **S3.7 Transfer-gap decomposition and the concept-shift diagnostic, in full**

For each direction, the gap between the within-region skill of the target and the raw transfer result is split in two parts. One part is what the best label-free adaptation recovers, and the other is what it does not recover. The recovered fraction is  $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$ , signed and not clipped, with its interval from the same paired bootstrap. It describes what the two methods tested achieved, with the better one chosen using target labels; other correction methods were not tested. Section 4.3 shows that the remainder should not be read as a

concept-shift residual, because much of it already appears inside a single region (Section S1.10).

Concept shift is examined with the **signed univariate association**. For each numeric predictor, the raw ROC-AUC of the predictor against burned is computed in each region and is not folded to  $\max(\text{AUC}, 1 - \text{AUC})$ , so a value below 0.5 shows a direction, not weakness. A reversal is called bootstrap-supported only when the point estimates of the two regions lie on opposite sides of 0.5 **and the interval of each region excludes 0.5**, under the same 10-cell spatial-block bootstrap. This is stricter than requiring disjoint intervals. A feature with disjoint intervals, one of which includes 0.5, has not been shown to point in any direction in that region, so it is recorded as a point reversal only. Section S2 repeats the rule next to the counts, and `conditional_similarity_transfer.json` stores it as metadata.

### S3.7.1 Degenerate replicates

Replicates with  $|G| < 10^{-6}$  are dropped.

## S3.8 Label-free adaptation, full specification

**Region-wise z-score.** The numeric features of each region are standardised with the statistics of that region: source statistics from source data and target statistics from target data, never pooled. For feature  $j$  in region  $R$ ,

$$z_{ij} = \frac{x_{ij} - \mu_j^R}{\sigma_j^R}, \quad \mu_j^R = \frac{1}{n_j^R} \sum_{i \in O_j^R} x_{ij}, \quad \sigma_j^R = \left( \frac{1}{n_j^R} \sum_{i \in O_j^R} (x_{ij} - \mu_j^R)^2 \right)^{1/2}, \quad (S4)$$

where  $O_j^R$  contains the  $n_j^R$  cells with an observed value (ddof 0). A missing value is first set to  $\mu_j^R$ , so it becomes zero, and  $\sigma_j^R < 10^{-12}$  is replaced by 1. Land cover is not transformed. Target feature statistics, but never target labels, are therefore used in the adapted models. The

classifier is refitted on the z-scored source and applied to the z-scored target. This removes differences in the mean and variance of each feature.

**CORAL after region-wise z-score.** The source covariance is aligned to the target covariance with the standard whitening and recolouring map (Sun et al. 2016). With  $Z_R$  the  $n_R \times d$  matrix of z-scored numeric features, one row per cell,

$$Z_s^{\text{al}} = Z_s (C_s + \lambda I)^{-1/2} (C_t + \lambda I)^{1/2}, \quad C_R = \frac{1}{n_R} \sum_{i=1}^{n_R} (z_i - \bar{z}_R)(z_i - \bar{z}_R)^\top, \quad (S5)$$

with  $\lambda = 10^{-5}$  (ddof 0). Both means are zero after Eq. (S4), so the mean terms of the general map are zero. Matrix powers use a symmetric eigendecomposition, with eigenvalues set to at least  $10^{-12}$ . **The transform is applied to the source only.** The target stays at  $Z_t$ , and the classifier is refitted on  $Z_s^{\text{al}}$ . Neither method uses a target label, and this is checked at run time. The sensitivity to  $\lambda$  was tested over nine values on four of the twenty directions; transfer AUC moved by at most 0.014, and  $\lambda$  was not selected on performance.  $\lambda = 1$ , the value of the original CORAL formulation, is tested in Sections S1.2 and S1.23. The value used throughout is  $\lambda = 10^{-5}$ .

## S4 Target-label recovery curve

### S4.1 Purpose

The main text shows that label-free alignment does not close the transfer gap. This section asks how much of the gap a small number of target labels can close. It is not a proposed method or an operational claim, because the labels come from the fire that is predicted. It covers three of the five regions (Manavgat, Bejís and Muğla) and all six directions among them. Source: the pipeline’s few-shot recovery diagnostic on the corrected label, analysis

7348dfe7..., released in paper/labelfix\_rerun/exports/few\_shot\_recovery\_7348dfe7/  
(recovery\_curve.csv, repeat\_metrics.csv, report.md); every value below is read from it.

## S4.2 Design

The unit of labelling effort is one 10-cell ( $\approx 5$  km) spatial block of the target. For a budget of  $k$  blocks,  $k$  target blocks are drawn from the training folds of a 5-fold spatially blocked split of the target. Their labels are added to the source training set, and the refitted model is scored on the held-out target folds. The budgets are 0, 1, 2, 4, 8, 16 and 32 blocks, with ten repeated draws for every  $k > 0$ . Budget 0 is raw transfer. The **ceiling** is a target-only model at the same 10-cell blocking (0.777 for Muğla, 0.824 for Bejís and 0.882 for Manavgat). It reproduces the pipeline's own large-block values exactly. The **recovered fraction** is  $(\text{few-shot} - \text{raw}) / (\text{ceiling} - \text{raw})$ , the share of the gap between raw transfer and the ceiling that the labels close; it is signed and not clipped. The intervals are **selection intervals**: the 2.5th and 97.5th percentiles over the ten block draws. They describe only which blocks were drawn, and no hypothesis test is made.

## S4.3 Result

**Table S22. Few-shot recovery of target ROC-AUC, thermal model, natural-vegetation population.** Raw = source-only transfer (budget 0); ceiling = target-only model at the same 10-cell blocking. Values are means over ten repeated block draws.

| Direction        | Raw   | 1 blk | 2 blk | 4 blk | 8 blk | 16 blk | 32 blk | Ceiling | Gap closed at 32 |
|------------------|-------|-------|-------|-------|-------|--------|--------|---------|------------------|
| Bejís → Manavgat | 0.314 | 0.518 | 0.565 | 0.659 | 0.698 | 0.750  | 0.820  | 0.882   | 89 %             |
| Muğla → Bejís    | 0.583 | 0.578 | 0.603 | 0.625 | 0.666 | 0.743  | 0.789  | 0.824   | 85 %             |
| Manavgat → Bejís | 0.396 | 0.447 | 0.456 | 0.541 | 0.598 | 0.690  | 0.752  | 0.824   | 83 %             |
| Muğla → Manavgat | 0.345 | 0.358 | 0.380 | 0.407 | 0.437 | 0.505  | 0.624  | 0.882   | 52 %             |
| Manavgat → Muğla | 0.438 | 0.447 | 0.453 | 0.468 | 0.490 | 0.532  | 0.571  | 0.777   | 39 %             |
| Bejís → Muğla    | 0.618 | 0.576 | 0.577 | 0.578 | 0.598 | 0.637  | 0.666  | 0.777   | 30 %             |

1120 **At 32 blocks, three of six directions close 83 to 89 % of the gap to the ceiling.** Two of  
1121 these three started below chance. The other three close 30 to 52 %. Thirty-two blocks contain  
1122 2,740 to 2,970 labelled cells, 7 to 20 % of the natural-vegetation population of the target. This  
1123 is not a small budget. For Bejís, the 32 labelled blocks contain on average 880 of its 1,100  
1124 burned cells, and for Manavgat about 1,800 of 2,935.

1125 **Recovery is slow where transfer is worst.** Muğla → Manavgat has the largest gap between  
1126 raw transfer and the ceiling. It is still below 0.5 after 8 labelled blocks and closes 52 % at 32.

1127 **Small budgets harm the direction that already transfers.** Bejís → Muğla is above chance  
1128 without labels (0.618), and target labels help it least. The curve is below the raw value at 1, 2,  
1129 4 and 8 blocks (0.576 to 0.598), rises above it only at 16 blocks, and closes 30 % at 32.

#### 1130 **S4.4 Limits**

1131 1. **Three regions, six directions.** Evia and Montiferru are not included, so this covers six  
1132 of the twenty directions and cannot cover all five regions. Six directions cannot  
1133 support a general label budget, so none is given.

1134 2. **It needs labelled target cells** from the fire that is predicted. Nothing here is a label-  
1135 free method, and nothing here changes the negative result for label-free alignment.  
1136 Labels from earlier fires in the same region were not tested.

1137 3. **The interval is a selection interval. At the largest budgets it becomes narrow**  
1138 **because the pool of blocks is used up, not because the value is well estimated.** Per  
1139 outer fold, the target has on average about 12 blocks with both classes in Bejís, 29 in  
1140 Manavgat and 48 in Muğla. At 16 and 32 blocks, the draws therefore use almost all of  
1141 these blocks and select almost the same set (`direction_budget_feasibility.csv`).

1142 4. **The ceiling is the 10-cell target-only value**, not the  $\approx 1$  km within-region value of  
1143 Table 1, so it is lower. The recovered fractions can only be read against this matched

1144 ceiling, and on the original study areas, which are not comparable across regions  
 1145 (Section 4.4).

## 1146 S5 Code

1147 The analysis code is public at <https://github.com/dryuemco/thermal-twin> under the MIT  
 1148 licence, and the version used for this paper is archived at Zenodo  
 1149 (<https://doi.org/10.5281/zenodo.23038039>). This section is a guide to it; the README of the  
 1150 repository gives the same information. The satellite processing pipeline is a separate public  
 1151 repository, <https://github.com/emrehann17/satellite-thermal-digital-twin> (MIT licence). The  
 1152 re-frozen Manavgat outputs were produced at its commit 6381f4c. The outputs of the other  
 1153 regions were produced at earlier commits and are reproduced by it (Section S3.6.2).

| Path                                       | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>paper/code/</code>                   | Analysis and verification scripts. <code>_canonical.py</code> loads every modelling dataset, verifies its SHA-256 and asserts the leakage exclusions of Section 3.13 at every fit.                                                                                                                                                                                                                                                                                                                         |
| <code>paper/code/appendix_tables.py</code> | Rebuilds Tables S2 to S7 and S9 to S18 row by row from source files whose SHA-256 values are pinned in <code>paper/labelfix_rerun/round5/tables/SOURCES.sha256</code> , and compares them with this document.                                                                                                                                                                                                                                                                                              |
| <code>paper/figures/</code>                | One script per figure. Each asserts every plotted value against the frozen outputs and the manuscript text, and checks its layout.                                                                                                                                                                                                                                                                                                                                                                         |
| <code>paper/figures/check_all.py</code>    | Runs all figure scripts and <code>appendix_tables.py</code> , restores the committed outputs byte for byte, and exits 0 when everything passes.                                                                                                                                                                                                                                                                                                                                                            |
| <code>paper/tex/</code>                    | Builds the manuscript and this document from their Markdown sources and checks the port: every number survives, and every cross-reference, citation and supplementary reference resolves.                                                                                                                                                                                                                                                                                                                  |
| <code>paper/labelfix_rerun/</code>         | The corrected-label outputs from which every corrected number is read: <code>code/</code> , <code>geometry/</code> , <code>inference/</code> , <code>labels/</code> and <code>round3/</code> to <code>round8/</code> hold the analysis outputs, <code>pipeline/</code> the re-frozen pipeline outputs for Manavgat, and <code>exports/</code> the released pipeline diagnostics (few-shot recovery, CORAL $\lambda$ sweep, window closure), each with its SHA-256 in <code>exports/SHA256SUMS.txt</code> . |

1154 All commands are run from the repository root. The environment is Python 3.12.10 with  
 1155 NumPy 2.4.4, pandas 3.0.2 and scikit-learn 1.9.0, fixed in `ENVIRONMENT.md`. Point estimates  
 1156 depend on the scikit-learn version (Section S3.5(vi)). Fig. 1 also needs cartopy; without it,  
 1157 `check_all.py` reports this figure as skipped. Tables S1, S8, S19 to S36 and the values in the

1158 text are not checked by `appendix_tables.py`; they name their source files in their captions or  
1159 text.

## 1160 **S6 Data**

1161 All satellite inputs are public and were retrieved through Google Earth Engine (Table S20).

1162 No proprietary or restricted data were used.

1163 The five modelling datasets are in the analysis repository, one per region, as  
1164 `paper/data/<region>/step8a_500m_modeling_dataset.parquet`. Each is checked on load  
1165 against the SHA-256 recorded in `paper/code/_canonical.py`. For Bejís, Muğla, North Evia  
1166 and Montiferru this hash equals the pipeline’s own record. For Manavgat the file is the  
1167 corrected-label re-freeze (SHA-256 5a5e876c...; Section S3.1.2). The pipeline’s record still  
1168 names the original-label table (SHA-256 054a1961...), which is not in the repository and can  
1169 be regenerated by the pipeline. The frozen numeric outputs behind every table and figure are  
1170 in the same repository, identified by the paths and hashes given in the table captions and in  
1171 `SOURCES.sha256`. The data derived from Landsat, MODIS, the Copernicus DEM and ESA  
1172 WorldCover remain subject to the terms of these products, which are given in the  
1173 Declarations.

## 1174 **References**

1175 Barve N, Barve V, Jiménez-Valverde A, et al (2011) The crucial role of the accessible area in  
1176 ecological niche modeling and species distribution modeling. *Ecological Modelling*  
1177 222:1810–1819. <https://doi.org/10.1016/j.ecolmodel.2011.02.011>

1178 Boschetti L, Roy DP, Giglio L, et al (2019) Global validation of the Collection 6 MODIS  
 1179 burned area product. *Remote Sensing of Environment* 235:111490.  
 1180 <https://doi.org/10.1016/j.rse.2019.111490>  
 1181 European Space Agency (2022) Copernicus DEM GLO-30 [dataset]. European Space  
 1182 Agency. <https://doi.org/10.5270/ESA-c5d3d65>  
 1183 Giglio L (2024) VIIRS/NPP burned area monthly L4 global 500m SIN grid V002 [dataset].  
 1184 NASA Land Processes Distributed Active Archive Center.  
 1185 <https://doi.org/10.5067/VIIRS/VNP64A1.002>  
 1186 Giglio L, Boschetti L, Roy DP, et al (2018) The Collection 6 MODIS burned area mapping  
 1187 algorithm and product. *Remote Sensing of Environment* 217:72–85.  
 1188 <https://doi.org/10.1016/j.rse.2018.08.005>  
 1189 Giglio L, Justice C, Boschetti L, Roy D (2021) MODIS/Terra+Aqua burned area monthly L3  
 1190 global 500m SIN grid V061 [dataset]. NASA Land Processes Distributed Active Archive  
 1191 Center. <https://doi.org/10.5067/MODIS/MCD64A1.061>  
 1192 Gorelick N, Hancher M, Dixon M, et al (2017) Google Earth Engine: Planetary-scale  
 1193 geospatial analysis for everyone. *Remote Sensing of Environment* 202:18–27.  
 1194 <https://doi.org/10.1016/j.rse.2017.06.031>  
 1195 Katagis T, Gitas IZ (2022) Assessing the accuracy of MODIS MCD64A1 C6 and FireCCI51  
 1196 burned area products in Mediterranean ecosystems. *Remote Sensing* 14:602.  
 1197 <https://doi.org/10.3390/rs14030602>  
 1198 Ludwig M, Moreno-Martinez A, Hölzel N, et al (2023) Assessing and improving the  
 1199 transferability of current global spatial prediction models. *Global Ecology and Biogeography*  
 1200 32:356–368. <https://doi.org/10.1111/geb.13635>

1201 Meyer H, Pebesma E (2021) Predicting into unknown space? Estimating the area of  
 1202 applicability of spatial prediction models. *Methods in Ecology and Evolution* 12:1620–1633.  
 1203 <https://doi.org/10.1111/2041-210X.13650>

1204 Meyer H, Pebesma E (2022) Machine learning-based global maps of ecological variables and  
 1205 the challenge of assessing them. *Nature Communications* 13:2208.  
 1206 <https://doi.org/10.1038/s41467-022-29838-9>

1207 Sandholt I, Rasmussen K, Andersen J (2002) A simple interpretation of the surface  
 1208 temperature/vegetation index space for assessment of surface moisture status. *Remote Sensing*  
 1209 *of Environment* 79:213–224. [https://doi.org/10.1016/S0034-4257\(01\)00274-7](https://doi.org/10.1016/S0034-4257(01)00274-7)

1210 Schoener TW (1968) The Anolis lizards of Bimini: Resource partitioning in a complex fauna.  
 1211 *Ecology* 49:704–726. <https://doi.org/10.2307/1935534>

1212 Sun B, Feng J, Saenko K (2016) [Return of frustratingly easy domain adaptation](#). In:  
 1213 *Proceedings of the thirtieth AAAI conference on artificial intelligence (AAAI-16)*. AAAI  
 1214 Press, Phoenix, Arizona, USA, pp 2058–2065

1215 U.S. Geological Survey Earth Resources Observation and Science (EROS) Center (2020)  
 1216 Landsat 8-9 operational land imager / thermal infrared sensor level-2, collection 2 [dataset].  
 1217 US Geological Survey. <https://doi.org/10.5066/P9OGBGM6>

1218 Wan Z, Hook S, Hulley G (2021) MODIS/Terra land surface temperature/emissivity daily L3  
 1219 global 1km SIN grid V061 [dataset]. NASA Land Processes Distributed Active Archive  
 1220 Center. <https://doi.org/10.5067/MODIS/MOD11A1.061>

1221 Warren DL, Glor RE, Turelli M (2008) Environmental niche equivalency versus  
 1222 conservatism: Quantitative approaches to niche evolution. *Evolution* 62:2868–2883.  
 1223 <https://doi.org/10.1111/j.1558-5646.2008.00482.x>

- 1224    Zanaga D, Van De Kerchove R, Daems D, et al (2022) ESA WorldCover 10 m 2021 v200.
- 1225    Zenodo. <https://doi.org/10.5281/zenodo.7254221>